

# Intensity of Volatility Linkages in Islamic and Conventional Markets

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## Abstract

Characteristics of Islamic finance, such as a smaller set of shared information and a lower degree of cross-market hedging, reduce volatility linkages (correlations) between Islamic and conventional stocks, bonds and bills. We use a stochastic volatility model in a Generalized Methods of Moments framework as well as other volatility proxies to estimate volatility linkages. We are the first to document that including at least one Islamic asset lowers volatility linkages by up to 7.17 percentage points, after controlling for country and asset-specific characteristics. Results are stronger during financial crises and are not driven by the oil sector.

*JEL Classification:* G1

*Keywords:* Volatility correlations; international markets; contagion; stochastic volatility; Islamic finance.

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# Intensity of Volatility Linkages in Islamic and Conventional Markets

## **Abstract**

Characteristics of Islamic finance, such as a smaller set of shared information and a lower degree of cross-market hedging, reduce volatility linkages (correlations) between Islamic and conventional stocks, bonds and bills. We use a stochastic volatility model in a Generalized Methods of Moments framework as well as other volatility proxies to estimate volatility linkages. We are the first to document that including at least one Islamic asset lowers volatility linkages by up to 7.17 percentage points, after controlling for country and asset-specific characteristics. Results are stronger during financial crises and are not driven by the oil sector.

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## 1. Introduction

While previous financial crises have been geographically limited, the recent US subprime crisis has spread to international financial markets. It has exerted a systemic impact that translated into global financial contagion, which in turn decreased the diversification benefits that investments would have experienced during normal economic times. This was due to financial markets having become more integrated over the past decade, which facilitated a faster propagation of the recent subprime US crisis to all sectors of the economy, as well as to other countries. The contagion effect translated into volatilities of most asset classes and most countries rising together. As a result, investors who held positions in different markets during the recent financial crisis found themselves exposed to more than one kind of volatility risk. Therefore, it has become crucial for investment and risk managers to assess the extent of the correlation between volatilities of different asset classes.

We are the first to examine this issue in an international context, while also exploring the way in which characteristics of Islamic financial markets affect the volatility linkages (correlations) between Islamic and conventional stock, bond and money market indices. We find that volatility linkages that involve at least one Islamic asset are lower than those between any two conventional assets by up to 13.53 percentage points, across all asset classes. For pairs of assets that include the Islamic equity index, the effect is stronger in Islamic relative to non-Islamic countries. Further, volatility linkages between stock, bond and money markets differ across countries due to the specific characteristics of their financial markets, as we document here. Our results are not driven by the oil sector. They hold during both crisis and non-crisis sub-periods, while being stronger during the former.

We estimate volatility linkages using: a) the Pearson correlation applied to both realized and *GARCH*-implied volatilities, and b) a stochastic volatility model estimated using

the *Generalized Method of Moments (GMM)*. We conduct both univariate and multivariate analyses to compare volatility linkages that involve one or two Islamic assets to volatility linkages between any two conventional assets. Throughout our analysis, we identify and control for various country and asset-specific factors that may affect volatility linkages.

Our data cover 9 Islamic and 37 non-Islamic countries for the period May 2007 to June 2010.<sup>1</sup> They include Islamic stock indices, conventional stock indices and conventional bond indices for all countries, and Islamic bond indices and conventional money market indices whenever they are available. Our study covers the most comprehensive dataset with Islamic assets, covering the largest number of countries for the longest period, subject to data availability.<sup>2</sup>

We find volatility linkages between Islamic stocks and bonds to be 12.72 to 86.77 percentage points lower than volatility linkages between conventional stocks and bonds in a univariate analysis, with the difference remaining significant and as large as 7.17 percentage points in a multivariate framework, where we control for country and asset-specific characteristics. Volatility linkages that involve only one Islamic asset (either stock or bond) are lower than those between conventional stocks and bonds, by up to 23.99 and 4.95 percentage points in the univariate and multivariate analysis, respectively. Volatility linkages between Islamic stocks and conventional bills are also lower than those between conventional stocks and bills, by up to 11.32 percentage points in a univariate analysis and by up to 2.51 percentage points in a multivariate analysis.

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<sup>1</sup> For the purpose of this paper, Islamic countries are those in which Islam is the main religion or that have an Islamic government.

<sup>2</sup> The sample period can no longer be extended because some data and variables that are required for the analysis are not available anymore.

We argue that the unique characteristics of Islamic stocks and bonds lower the volatility linkages by reducing the set of common information and the degree of cross-market hedging, which are the two key drivers of volatility linkages across asset classes. Islamic finance practices socially responsible investment, and requires compliance with Islamic law (*Shari'ah*). This law prohibits interest or usury (*riba*); transactions involving unnecessary uncertainty or a deliberate lack of transparency (*gharar*); and gambling (*qimar*), which covers short selling, arbitrage and speculation. Instead, financial arrangements that include Islamic bonds (*sukuk*) are based on profit-and-loss sharing, with an additional emphasis on the link between the real and the financial sectors. Further, Islamic stock indices exclude certain 'unethical' sectors and firms that derive significant income from interest, or that incur a large amount of leverage.

The difference in volatility linkages that we document between bad and good economic times is indicative of the recent spread of financial contagion across global markets. Our results have important economic implications, as financial crises are characterized by a strong flight to quality whereby, for instance, investors shift their preferences from stocks to bonds in the expectation that stock market volatility will increase. By doing so, they are attempting to reduce their stock market risk exposure. However, whether they achieve this depends on the extent to which the volatility of one market is linked to the volatility of another. If volatility linkages are stronger during bad economic times, then bonds may not provide the safe haven that investors are seeking.

Our research is timely, as there has been an increased interest in the transmission of global financial contagion (Bekaert et al., 2005; Bekaert et al., 2009; Aït-Sahalia and Hurd, 2012; Adams et al., 2014; Bekaert et al., 2014; Bongaerts et al., 2014; Aït-Sahalia et al., 2015; Jawadi et al., 2015; among others). Our findings have important implications when

devising international investment allocations and risk management strategies, especially since there has been a rising interest in Islamic finance. This is one of the fastest growing segments of the global financial services industry, with Islamic financial assets having grown at a substantial average annual growth of 17.6 per cent over 2008–2012, increasing 50 per cent faster than the overall banking sector (Ernst & Young, 2012; 2013). Global Islamic financial assets were \$1.7 trillion in 2013 and they are forecast to grow to \$3.4 trillion by 2018 (Ernst & Young 2013). While the countries of the core Middle East (i.e., Iran, Gulf States), Malaysia and Indonesia still represent fertile ground for future growth, the industry has expanded beyond these markets and now covers many Western countries. For example, Islamic financial services have been established by several Western banks and financial institutions, such as Citigroup, HSBC, Barclays and UBS (Mehyo, 2008), and Islamic indices have been created by prominent providers such as MSCI Barra, Dow Jones, FTSE and Standard and Poor's. The Australian Board of Taxation, for instance, initiated a review of taxation laws to ensure that the system does not disadvantage or preclude Islamic instruments and that it supports the growth of Islamic finance in Australia (Australian Trade Commission, 2010).<sup>3</sup>

The remainder of this paper is structured as follows. Section 2 reviews the literature relevant to our research hypothesis and Islamic finance principles. Section 3 presents the methodology used to test our research hypothesis. Section 4 describes the data. Section 5 provides the results for both the univariate and multivariate analyses. Section 6 performs robustness tests. Section 7 concludes with a summary of the main findings and their practical implications. Appendix A summarizes the data availability for each country, and specifies the

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<sup>3</sup> Further, expansion of Islamic assets is expected in countries such as Australia, Azerbaijan, Nigeria, Russia, Hong Kong, Japan, Singapore and United Kingdom (Australian Trade Commission, 2010). Morocco is set to launch its first fully-fledged Islamic Bank by late 2015 according to Bloomberg News (March 2015).

names and data sources for our stock, bond and money market indices. It also documents the sources of our control variables. Appendix B provides annualized returns and volatilities for a subset of countries, and presents additional robustness tests for sub-samples of countries for which we have the necessary data.

## **2. Review of Volatility Linkages and Hypothesis Development**

### *2.1. Related Literature*

Accounting for volatility correlations between asset classes is essential for portfolio and risk managers, derivative dealers and policy regulators, and various studies have focused on estimating them within a given country (Fleming et al., 1998; Fleischer, 2003; Antell, 2004; Chulia and Torro, 2008; Nguyen et al., 2014). As the past decade has seen an increase in financial market interdependence, the literature has also looked at cross-market rebalancing and its relation to financial market contagion (Kodres and Pritsker, 2002; Bekaert et al., 2014), and at volatility spillovers across stock markets (King et al., 1994; Forbes and Rigobon, 2002; Baele, 2005; Diebold and Yilmaz, 2009; Engle et al., 2012; Zhou et al., 2012; Abidin et al., 2014; Majdoub and Mansour, 2014; Bhuyan et al., 2015).

Volatility linkages can be modelled as the information linkages between two markets, based on the proportional relation between volatility and information flow. Since information flow is unobservable, we use the speculative trading model developed by Fleming et al. (1998), which is based on the trading model set in a rational expectations framework by Tauchen and Pitts (1983), and on the work of Clark (1973), Ross (1989) and Andersen

(1996).<sup>4</sup> While Tauchen and Pitts (1983) consider a single futures contract, Fleming et al. (1998) generalize the rational expectations model by letting investors trade in more than one futures market, and then estimate the volatility linkages between the stock, bond and money markets via a stochastic volatility model. We do not focus on futures contracts here, as derivatives remain controversial in most Islamic countries, and those that are traded are either based on commodities with actual physical delivery or are explicitly not compliant with Islamic law (Zaher and Hassan, 2001; Jobst, 2007). Instead, we focus on the spot markets for stock, bond and money. In doing so, we build on the literature on volatility and information linkages, while also extending it to include Islamic assets.

The first source for volatility linkages across asset classes is the set of common information, where macroeconomic news and information simultaneously affect expectations in more than one market. The second source is the information spillover caused by cross-market hedging. The latter occurs when investors operate in more than one asset class and thus respond to shocks in one market by optimally readjusting their portfolios in the other markets (Kodres and Pritsker, 2002).<sup>5</sup>

## *2.2. Islamic Finance*

There are several reasons why volatility linkages in Islamic markets may differ from those recorded in conventional markets. First, Islamic principles can affect both of the drivers of

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<sup>4</sup> Tauchen and Pitts (1983) assume a large number of active speculators who trade due to differences in expectations about the future. The market is in equilibrium at the beginning of each trading day and the investors revise their expectations and trade as new information arrives. Further, in that model, the variance in daily returns is directly proportionate to daily information flow.

<sup>5</sup> As a result, there is a price movement in one market due to shocks in another market. As a side note, cross-market rebalancing has been a major factor in the recent financial contagion across international markets.



volatility linkages; that is, the set of common information and the degree of cross-market hedging. The prohibition of interest or usury (*riba*) and the strong emphasis on the performance of underlying assets in determining the payoff to investors have led to a smaller set of information common across Islamic securities (or across Islamic and conventional securities) than that common across conventional securities.<sup>6</sup> For example, the set of macroeconomic information characteristic to Islamic markets includes factors such as inflation and GDP, but not interest rates. Hakim and Rashidian (2002) show, for instance, that the Dow Jones Islamic Index in the US has no stable link with the Wilshire 5000 Index or the T-bill rate, even though the Islamic index is 100% composed of US stocks. This suggests that the Islamic index is influenced by an entirely different set of factors (e.g., the personal income of *Shari'ah*-conscientious investors). Further, cross-asset hedging tends to be lower in Islamic markets due to the characteristics of Islamic finance, such as the prohibition of speculation (*gharar*); short selling, betting and gambling (*qimar*); and arbitrage (Jobst, 2007). Another contributing factor is the weak secondary debt market in most Islamic countries (Tariq and Dar, 2007).

Second, there are structural differences between Islamic and conventional assets. Islamic bonds (*sukuk*) are investment certificates that are based on various Islamic partnership and leasing arrangements.<sup>7</sup> However, all of them are backed by tangible assets and most of the Islamic bonds are independent of interest rate movements, such that the profit

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<sup>6</sup> See for example El-Gamal (2003) and Jobst (2007) for a more detailed discussion of the characteristics of Islamic finance and the alternative financial products that comply with Islamic law.

<sup>7</sup> For example, within a leasing arrangement, an Islamic bond is structured such that the investor funds the firm's asset and then receives regular rent payments until the maturity of the contract, when the firm buys back the asset.

depends on the performance of the underlying asset.<sup>8</sup> An example of how investors perceive *sukuk* and conventional bonds differently is a Malaysian study by Godlewski et al. (2013) who show that there is no significant stock market reaction when an announcement of a conventional bond issue is made, whereas the stock market reacts negatively to a *sukuk* issue. The authors argue that this is due to excess demand for *sukuk* by Islamic financial institutions, as well as an adverse selection mechanism which encourages companies with low profit expectations to prefer issuing *sukuk* over conventional bonds. Likewise, Alam et al. (2013) study announcements of bond issues in Malaysia, Indonesia, Singapore, Pakistan, UAE, Bahrain and Qatar, and confirm the finding in Godlewski et al. (2013) regarding *sukuk* during crisis and non-crisis periods, although they find a positive (negative) stock market reaction for conventional bond issues before (after) crisis periods.

As for equity, several adjustments need to be made to conventional stock indices to ensure their compliance with *Shari'ah* law. As such, Islamic stock indices are a subset of conventional stock indices and are constructed in two steps: 1) stocks of companies whose business is in areas not suitable for Islamic investment purposes are excluded from the index (such as alcohol, tobacco, conventional banks and insurance companies, gambling and entertainment industries), and 2) financial ratio filters are applied to exclude firms with large amounts of debt and interest income.<sup>9</sup> Therefore, a major distinction between Islamic and

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<sup>8</sup> Islamic bonds must not take advantage of interest-rate movements. However, critics have pointed out that not all Islamic bonds comply with Islamic law, as for some of them the profit is tied to LIBOR or other rates (see Tariq and Dar, 2007 and Alaoui et al., 2015).

<sup>9</sup> See, for example, Zaher and Hassan (2001), Derigs and Marzban (2009) and Ho et al. (2014) for an overview of business activity and financial ratio screens. Screening criteria vary slightly across providers. For example, Dow Jones (2011) applies business screens to exclude firms from the following sectors: Alcohol, Tobacco, Pork, Conventional financial services (except for financial companies that are incorporated as an Islamic Financial Institution, such as Islamic Banks or Takaful Insurance Companies, Weapons and Defence, and Entertainment. The second set of screens are financial ratio filters:

conventional stock indices is that they reflect different samples of industries and firms. For example, Forte and Miglietta (2007) draw comparisons between the FTSE Islamic index, the conventional FTSE Developed Europe and the SR FTSE4 Good (socially responsible index) within Europe, finding that the Islamic index shows unique characteristics due to the inclusion of certain industries, such as oil and gas industries, and the exclusion of other industries, like conventional financial companies.<sup>10</sup> Majdoub and Mansour (2014) find that among Islamic stock indices there is only a low level of volatility spillover from the US stock market to Islamic emerging stock markets (Indonesia, Malaysia, Pakistan, Qatar, Turkey). They attribute this finding to features of Islamic finance, such as the exclusion of volatile sectors (e.g. gambling) from the Islamic stock indices, the limited exposure to the volatility of interest rates, as well as the asset-backed principle which encourages a close linkage between the real and financial sectors and discourages speculative investments (e.g. derivatives and short-selling).

In sum, the characteristics of Islamic bonds and stocks are such that they include firms with low leverage, from selected industries, that have close ties to the underlying assets.

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Leverage compliance (Total debt divided by trailing 24-month average market capitalisation < 33%), and Cash Compliance (The sum of a company's cash and interest-bearing securities divided by trailing 24-month average market capitalisation < 33% and accounts receivables divided by trailing 24-month average market capitalisation < 33%). While there is some resemblance between Islamic business activity screens and screens of other ethical (or socially responsible) stock indices, there is a major difference as Islamic indices also exclude other prohibited business sectors such as conventional banks and insurance companies, as well as firms with high levels of debt, cash and interest-bearing securities or receivables. This results in entirely different risk and return profiles between Islamic and ethical stocks, or alternatively between non-Islamic and sin stocks (Jahromi, 2014).

<sup>10</sup> Financial institutions with a large exposure to Islamic finance may benefit from Islamic loans boasting a default rate that is less than half the default rate of conventional loans, as shown by Baele et al. (2014) in a sample of 150,000 loans in Pakistan. They also show that a borrower who concurrently holds Islamic and conventional loans is less likely to default on the Islamic loan.

This could affect the volatility linkages through different risk profiles, regulations and a stronger dependency of volatility on the firm's performance, rather than through common macroeconomic factors like interest rates.<sup>11</sup> Based on the different structure of Islamic stock and bond indices, as well as on the impact of Islamic principles on the set of common information and cross-assets hedging, our main hypothesis is that volatility linkages involving Islamic indices are lower than those involving conventional indices. We test our hypothesis in the subsequent section using data on Islamic and conventional assets, collected from both Islamic and non-Islamic countries. We start by presenting our methodology for deriving the relation between volatility and information flow.

### 3. Methodology

We estimate volatility linkages per asset pair by using data across all years of our sample period, and then test whether there is a significant difference between these linkages across markets and countries. Comparisons between volatility linkages are drawn between different combinations of Islamic and conventional assets, in both Islamic and non-Islamic countries. We use two approaches to estimate the volatility linkages within each country: (i) the Pearson correlation and (ii) a stochastic volatility model. Under the first approach, we compute the volatility linkages for different pairs of assets as the Pearson correlations of the respective volatility series. We use daily data to build daily volatility series, which we proxy with the daily series of absolute returns,  $|r_{k,t}| = \left| \ln \left( \frac{P_{k,t}}{P_{k,t-1}} \right) \right|$ , or the daily series of squared returns,

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<sup>11</sup> An example for how leverage may affect volatility linkages is given in Kodres and Pritsker (2002), who argue that a dramatic increase in financial leverage of some companies during an exchange rate or financial crisis can increase risk factor sensitivities and, as such, the volatility linkages between countries.

$$(r_{k,t})^2 = \left( \ln \left( \frac{P_{k,t}}{P_{k,t-1}} \right) \right)^2, \text{ for each asset } k \text{ (Pagan and Schwert, 1990; West and Cho, 1995).}^{12}$$

Under the second approach, we apply the stochastic volatility model detailed below, while generating restrictions on the proportional relation between the information flow and volatility to estimate volatility linkages.<sup>13</sup> We are the first to apply this stochastic volatility model within the framework of Islamic and conventional assets to examine how volatility linkages vary across these asset types.

### 3.1. Stochastic Volatility Model

The trading model in Fleming et al. (1998) assumes that the daily information flow is stochastic. It is proportional to the variance of daily returns and thus generates volatility. In that model, the unpredictable component of returns is due to the arrival of new information. Specifically, let  $R_{k,t}$  be the return and  $\mu_{k,t}$  be the conditional expected return for asset class  $k$  on day  $t$ . We assume that asset class  $k$  is affected by  $I_{k,t}$  information events during day  $t$ . Let  $\varepsilon_{ik,t}$  be the day  $t$  shock to the return of asset  $k$ , due to event  $i$  ( $i=1 \dots I_{k,t}$ ). We assume the shocks to be i.i.d. normally distributed:  $\varepsilon_{ik,t} \sim N(0, \sigma_{\varepsilon,k}^2)$ , and write the daily return as the sum between the expected and unexpected components:

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<sup>12</sup> Alternatively, forward looking volatility proxies could be used, such as the VIX, which is implied by the current prices of options on the S&P 500 Index. However, the construction of an implied volatility index requires a deep and active index option market of the underlying index (Whale, 2009). For this reason, implied volatility indices cannot be used as volatility proxies in this study due to the lack of deep and active underlying index option markets in most countries, and particularly in the case of Islamic stock and bond indices.

<sup>13</sup> Various complex models have been developed to capture volatility, for instance Aït-Sahalia and Jacod (2007) and Aït-Sahalia et al. (2011). These models are highly sophisticated from a theoretical point of view and superb for high frequency data. However, given the daily frequency of our data, we do not consider these methods here.

$$R_{k,t} = \mu_{k,t} + \sum_{i=1}^{I_{k,t}} \varepsilon_{ik,t} . \quad (1)$$

The latter component of returns is composed of all the intra-daily shocks that are due to the arrival of new information. Daily returns are thus the sum of incremental intra-daily returns associated with news events. Eq. (1) can be rewritten as:

$$R_{k,t} = \mu_{k,t} + \sigma_{\varepsilon,k} \sqrt{I_{k,t}} z_{k,t} , \quad (2)$$

where

$$z_{k,t} = \frac{1}{\sqrt{I_{k,t}}} \sum_{i=1}^{I_{k,t}} \left( \frac{\varepsilon_{ik,t}}{\sigma_{\varepsilon,k}} \right) \xrightarrow{I_{k,t} \rightarrow \infty} N(0,1) . \quad (3)$$

The result in (3) is obtained using the Central Limit Theorem. Eqs. (2) and (3) imply that the conditional distribution of  $R_{k,t}$  should be approximately  $N(\mu_{k,t}, \sigma_{\varepsilon,k}^2 I_{k,t})$ . This tells us that  $\text{Var}(R_{k,t}) \propto I_{k,t}$ . More precisely:

$$\sigma(R_{k,t}) = \sigma_{\varepsilon,k} \sqrt{I_{k,t}} . \quad (4)$$

Eq. (4) shows the actual link between information flow and volatility: volatility of asset class  $k$  is stochastic and depends on the information occurring during day  $t$ .

Using the usual notation from the literature, let  $h_{k,t} = \ln(\sigma_{\varepsilon,k}^2 I_{k,t})$  be the log daily variance following an  $AR(1)$  specification:

$$h_{k,t} = \phi_{k,0} + \phi_{k,1} h_{k,t-1} + u_{k,t} , \quad (5)$$

where  $u_{k,t}$  are i.i.d. shocks to volatility, with mean zero and variance equal to  $\sigma_{u,k}^2$ , and independent of  $z_{k,t}$ . We assume that  $|\phi_{k,1}| < 1$ , for the variance process to be stationary. Eqs. (2) and (5) imply that the day  $t$  returns for asset class  $k$  are generated by the joint stochastic process:

$$\begin{cases} R_{k,t} = \mu_{k,t} + \sigma_{\epsilon,k} \sqrt{I_{k,t}} z_{k,t} = \mu_{k,t} + (\exp(h_{k,t}))^{1/2} z_{k,t} \\ h_{k,t} = \phi_{k,0} + \phi_{k,1} h_{k,t-1} + u_{k,t} \end{cases} . \quad (6)$$

### 3.2. Moment Restrictions

We now focus on the unexpected component of returns:

$$r_{k,t} = R_{k,t} - \mu_{k,t}, \quad (7)$$

which is a stationary process (being the product of two stationary processes). Substituting Eq. (7) into system (6) gives:

$$r_{k,t} = (\exp(h_{k,t}))^{1/2} z_{k,t}, \quad (8)$$

or equivalently:

$$\ln(r_{k,t}^2) = h_{k,t} + \ln(z_{k,t}^2). \quad (9)$$

We denote by:

$$y_{k,t} = \ln(r_{k,t}^2) - E[\ln(z_{k,t}^2)] . \quad (10)$$

As a result, system (6) becomes:

$$\begin{cases} y_{k,t} = \ln(r_{k,t}^2) - E[\ln(z_{k,t}^2)] = h_{k,t} + \{\ln(z_{k,t}^2) - E[\ln(z_{k,t}^2)]\} = h_{k,t} + \xi_{k,t} \\ h_{k,t} = \phi_{k,0} + \phi_{k,1} h_{k,t-1} + u_{k,t} \end{cases} \quad (11)$$

Since  $z_{k,t} \sim N(0,1)$ , the mean and variance of  $\ln(z_{k,t}^2)$  are -1.27 and 4.93, respectively (Abramowitz and Stegun, 1970). Therefore:

$$\xi_{k,t} = \ln(z_{k,t}^2) - E[\ln(z_{k,t}^2)] \quad (12)$$

has mean zero and variance 4.93 (Fleming et al., 1998).  $\xi_{k,t}$  is independent of  $h_{k,t}$ . Further, as stated above,  $h_{k,t}$  is covariance stationary. Thus:

$$\mu_{h,k} = E[h_{k,t}] = \frac{\phi_{k,0}}{1-\phi_{k,1}} \quad (13)$$

and

$$Var(h_{k,t}) = \frac{\sigma_{u,k}^2}{1-(\phi_{k,1})^2}. \quad (14)$$

Next, we compute the first two moments of  $y_{k,t}$  in (11) and obtain:

$$\begin{cases} E[y_{k,t}] = E[h_{k,t}] \\ Var(y_{k,t}) = Var(h_{k,t}) + Var(\xi_{k,t}) \\ Cov(y_{k,t}, y_{k,t+\tau}) = (\phi_{k,1})^\tau \frac{\sigma_{u,k}^2}{1-(\phi_{k,1})^2} = (\phi_{k,1})^\tau Var(h_{k,t}) \end{cases} \quad \text{for all integers } \tau > 0. \quad (15)$$

We build a bivariate generalization of system (11) to estimate the cross-markets volatility linkages:

$$\begin{cases} \begin{cases} y_{i,t} = h_{i,t} + \xi_{i,t} \\ y_{j,t} = h_{j,t} + \xi_{j,t} \end{cases} \\ \begin{cases} h_{i,t} = \phi_{i,0} + \phi_{i,1}h_{i,t-1} + u_{i,t} \\ h_{j,t} = \phi_{j,0} + \phi_{j,1}h_{j,t-1} + u_{j,t} \end{cases} \end{cases} \quad (16)$$

Consistent with the above assumptions,  $[h_{i,t}, h_{j,t}]'$  is independent of  $[\xi_{i,t}, \xi_{j,t}]'$ . Note that the correlation between  $h_{i,t}$  and  $h_{j,t}$  measures the volatility linkage between markets  $i$  and  $j$ . In the case that the information spillover between any two given markets is complete, the correlation between their corresponding volatilities should be equal to one.

Similar to system (15), and based on the assumptions stated above, we now construct a set of cross-markets moments for system (16):



$$\begin{cases} Cov(y_{i,t}, y_{j,t}) = Cov(h_{i,t}, h_{j,t}) + Cov(\xi_{i,t}, \xi_{j,t}) \\ Cov(y_{i,t}, y_{j,t+\tau}) = (\phi_{j,1})^\tau Cov(h_{i,t}, h_{j,t}) \\ Cov(y_{i,t+\tau}, y_{j,t}) = (\phi_{i,1})^\tau Cov(h_{i,t}, h_{j,t}) \end{cases}, \text{ for all } \tau > 0 \text{ integers. (17)}$$

### 3.3. GMM Estimation

We estimate volatility linkages using the *GMM* and the relation between information flow and volatility derived above. First, we seasonally adjust both the returns and volatility series of each asset class. The seasonal patterns in returns are removed by regressing them on five weekday dummy variables and one dummy variable for the days following market holidays. The seasonal patterns in volatility are then removed by regressing the log of the residuals squared from the previous regression on a dummy variable for Mondays and a dummy variable for days following market holidays. We then subtract  $E[\ln z_{k,t}^2]$  from the intercept and residuals of this regression to obtain the seasonally adjusted series  $y_{k,t}$  from Eq. (10):

$$y_{k,t} \equiv \ln r_{k,t}^2 - E[\ln z_{k,t}^2]. \quad (18)$$

Using the moment restrictions (15) and (17), we next build the corresponding *GMM* disturbance vectors. This is done for each individual asset  $k$  and for bivariate systems across different asset classes within one country. The univariate estimations involve the moment restrictions for each individual asset  $k$ , for both Islamic and conventional assets, in Islamic and non-Islamic countries. The *GMM* disturbance vector corresponding to system (15) can be expressed as:

$$e_t(\theta) = \begin{pmatrix} y_{k,t} - \mu_{h,k} \\ (y_{k,t} - \mu_{h,k})^2 - \sigma_{h,k}^2 - \sigma_\xi^2 \\ (y_{k,t} - \mu_{h,k})(y_{k,t+\tau} - \mu_{h,k}) - (\phi_{k,1})^\tau [(y_{k,t} - \mu_{h,k})^2 - \sigma_\xi^2] \end{pmatrix}, \quad (19)$$

where  $\theta \equiv [\mu_{h,k}, \sigma_{h,k}^2, \phi_{k,1}]'$  is the vector of unknown parameters, and  $\tau = 1, 2, \dots, L$  counts the number of autocorrelation restrictions used in the estimation. The first two restrictions identify the mean ( $\mu_{h,k}$ ) and variance ( $\sigma_{h,k}^2$ ) of the log information flow  $h_{k,t}$ , and the  $L$  remaining restrictions identify the  $AR(1)$  parameter  $\phi_{k,1}$  of the  $h_{k,t}$  process.

The cross-market correlations are estimated by fitting various bivariate systems. The pairings involve different asset classes within one country, such as Islamic and conventional stock, bond and money market assets. The bivariate systems combine the univariate disturbance vectors for markets  $i$  and  $j$  together with the disturbance vector based on system (17):

$$e_t(\theta_i, \theta_j) = \begin{pmatrix} (y_{i,t} - \mu_{h,i})(y_{j,t} - \mu_{h,j}) - \rho_{h,ij}\sigma_{h,i}\sigma_{h,j} - \sigma_{\xi}^2\rho_{\xi,ij} \\ (y_{i,t} - \mu_{h,i})(y_{j,t+\tau} - \mu_{h,j}) - (\phi_{j,1})^\tau[(y_{i,t} - \mu_{h,i})(y_{j,t} - \mu_{h,j}) - \sigma_{\xi}^2\rho_{\xi,ij}] \\ (y_{i,t+\tau} - \mu_{h,i})(y_{j,t} - \mu_{h,j}) - (\phi_{i,1})^\tau[(y_{i,t} - \mu_{h,i})(y_{j,t} - \mu_{h,j}) - \sigma_{\xi}^2\rho_{\xi,ij}] \end{pmatrix} \quad (20)$$

where  $\theta_i \equiv [\mu_{h,i}, \sigma_{h,i}^2, \phi_{i,1}]$  represents the vector of unknown parameters for market  $i$ , and  $\theta_j \equiv [\mu_{h,j}, \sigma_{h,j}^2, \phi_{j,1}]$  corresponds to market  $j$ . The  $i$  and  $j$  subscripts denote the two asset classes across which the volatility linkages are estimated. These asset classes can be Islamic or conventional and refer to either stock, bond or money markets. Further, each bivariate system includes two additional parameters:  $\rho_{h,ij}$ , which represents the correlation between the information flow across asset classes; and  $\rho_{\xi,ij}$ , which provides the correlation between the disturbance terms in markets  $i$  and  $j$ . As such, the estimated cross-market linkages can be obtained from the correlation of the log information flows between markets,  $\widehat{\rho_{h,ij}}$ , which reflects the strength of the volatility linkages. Combining the univariate and bivariate systems, the joint system has  $4L+5$  equations, with eight unknown parameters. The overidentifying  $J$  statistic has the *Chi squared* distribution  $\chi_{(4L+5)-8}^2$ .

### 3.4. GARCH Model

In our robustness section, we compute the Pearson correlation between volatilities implied by a *Generalized Autoregressive Conditional Heteroskedastic (GARCH)* model applied to the individual time series of asset returns. The *GARCH* model is specified as follows.

Let  $y_t$  be a return series with the conditional distribution  $y_t|\Theta_{t-1} \sim N(0, h_t)$ , where  $\Theta_{t-1}$  denotes all the available information at time  $t-1$ . The conditional variance  $h_t$  follows a *GARCH(p,q)* specification:

$$h_t = \alpha_0 + \sum_{i=1}^q \alpha_i y_{t-i}^2 + \sum_{j=1}^p \beta_j h_{t-j}, \quad (21)$$

where  $p \geq 0, q > 0, \alpha_0 > 0, \alpha_i \geq 0, \beta_j \geq 0$ .

The individual *GARCH* models are estimated using the maximum likelihood method, where the log-likelihood function is computed using the product of all conditional densities of the prediction errors. We determine the order of the fitting *GARCH* models for all of the time series of returns based on the Akaike and Schwarz Bayesian information criteria, and by examining the model residuals.

### 3.5. Testing our Hypothesis

As mentioned above, both univariate and multivariate analyses are employed to test the research hypothesis that volatility linkages involving Islamic assets are lower than those involving conventional assets. The univariate analysis involves creating a series of differences between series (i) and (ii), where (i) represents volatility linkages across conventional assets, and (ii) represents volatility linkages involving at least one Islamic asset. For example, one difference term is the volatility linkage between conventional US stocks

and US bonds, minus the volatility linkage between Islamic US stocks and conventional US bonds. In this way, we obtain 96 distinct difference terms. We then conduct t-tests on the hypothesis that the difference terms are positive; that is, volatility linkages involving any two conventional assets are stronger than those involving at least one Islamic asset.

While the univariate analysis is based on our estimates of volatility linkages across all assets (that is, we have one estimate per asset pair within each country), our multivariate tests are meant to capture the change in volatility linkages across time, while controlling for additional factors that may affect the linkages. To this end, the multivariate analysis consists of estimating the following regression:

$$\begin{aligned} \rho_{i,a,b,t} = & \beta_0 + \beta_1 I_{\{Islamic\ Country\}} + \beta_2 I_{\{Islamic\ Stock\ and\ Islamic\ Bond\}} + \\ & \beta_3 I_{\{Islamic\ Stock\ and\ Conventional\ Bond\}} + \beta_4 I_{\{Islamic\ Bond\ and\ Conventional\ Stock\}} + \\ & \sum_{m=6}^n \beta_{i,m} x_{i,m} + \varepsilon_{i,t} , \end{aligned} \quad (22)$$

where

$I_{\{Islamic\ Country\}}$ : Dummy variable that takes the value 1 if the volatility linkage is measured in an Islamic country and 0 otherwise.

$I_{\{Islamic\ Stock\ and\ Islamic\ Bond\}}$  through  $I_{\{Islamic\ Bond\ and\ Conventional\ Stock\}}$  : Dummy variables that take the value 1 if the volatility linkage is measured across at least one Islamic asset. The control group is the case in which the volatility linkage is estimated between two conventional assets (stocks and bonds).

$x_{i,m}$  : Control variables that are intended to capture other factors that may affect volatility linkages through time and across countries. Details are provided in Section 4.2 below and in Table A.4 of Appendix A.

The dependent variable  $\rho_{i,a,b,t}$  is the monthly volatility linkage in country  $i$  at time  $t$ . It is measured as the monthly correlation between the daily return volatilities across any two assets  $a$  and  $b$ , where asset  $a$  is the Islamic or conventional stock index, and asset  $b$  is the Islamic or conventional bond index. The reason for using monthly correlations rather than correlations across the whole sample period (as in our univariate analysis) is that there is considerable variation in volatility linkages across time, which the multivariate analysis is intended to capture.

We repeat the exercise in Eq. (22) by applying it to the stock and money markets. To this end, we modify Eq. (22) by defining asset  $b$  as the money market index instead of the bond index. Note that there are no Islamic money market indices. We obtain:

$$\rho_{i,a,b,t} = \beta_0 + \beta_1 I_{\{Islamic\ Country\}} + \beta_2 I_{\{Islamic\ Stock\ and\ Money\ Market\}} + \sum_{m=6}^n \beta_{i,m} x_{i,m} + \varepsilon_{i,t} . \quad (23)$$

where

$I_{\{Islamic\ Stock\ and\ Money\ Market\}}$ : Dummy variable that takes the value 1 if the volatility linkage is measured between Islamic stock and conventional money market. The control group is the case in which the volatility linkage is estimated between conventional stock and conventional money market.

## 4. Data

### 4.1. Data Collection and Adjustments

We collect daily return data on conventional stock, bond and money market indices for 9 Islamic countries, 37 non-Islamic countries and a world index. Daily frequency is the highest level for which we can obtain data. We select our sample such that it includes the largest

number of countries over the longest period, subject to data availability. As such, each country in our sample has data on a conventional and an Islamic stock index, a conventional and an Islamic bond index, and a conventional money market index, when available. Note that Islamic bond indices are only available in Malaysia, the United Arab Emirates (UAE), the Gulf Cooperation Council (GCC) and as a world index. The sample period is from 31 May 2007 to 8 June 2010. Table A.1 of Appendix A summarizes the data availability for each country, while Tables A.2 and A.3 of the same Appendix list all the names and data sources for the stock, bond and money market indices.

To make volatilities comparable across countries, all return series are converted into US dollars using the corresponding daily exchange rate obtained from DataStream. Then, we match the dates of different assets within each country, excluding the corresponding public holidays. Finally, a daily return series is created for each asset  $k$  as the log of price relatives:

$$r_{k,t} = \ln\left(\frac{P_{k,t}}{P_{k,t-1}}\right). \text{ We also estimate the 10-lag autocorrelations for each daily return series.}$$

However, we do not find any apparent pattern within the autocorrelation structure across countries or market types.<sup>14</sup> For illustration purposes, Table B.1 of Appendix B provides the number of observations and the annualized returns and volatilities for a subset of countries included in this study. In many instances, the stock series have not provided a higher return compared to the bond and bill series, and sometimes the stock returns have even been negative during our sample period. This could be due to the adverse effects of the global financial crisis on stock markets in particular. As expected, the standard deviation is mostly highest for the stock series, followed by corporate bonds and then Treasury bonds and bills.

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<sup>14</sup> Results are not reported in the interest of brevity, but they are available upon request.

#### *4.2. Control Variables*

We use control variables to capture any factors that may affect volatility linkages over time and across countries. According to Fleming et al. (1998), volatility linkages are determined by the set of common information that simultaneously affects expectations across financial markets, and by information spillovers caused by cross-market hedging. The set of information common to the stock, bond and bill indices depends on the underlying assets. For example, a bond index may include government or corporate bonds, and bonds with different maturities. These characteristics are controlled for by including an indicator variable for corporate bonds in Eqs. (22) and (23). Several variables are also included to capture the degree of market uncertainty that could drive the sensitivity of stocks, bonds and bills to common information, such as sovereign credit ratings, country risk, periods of high or low volatility and countries' GDP per capita.<sup>15</sup> Further, various factors that affect the ease of cross-market hedging have been proposed in theoretical and empirical studies, such as regulations (including transaction costs, institutional constraints and information barriers) and market liquidity.<sup>16</sup> We proxy for institutional constraints and information barriers in this study with items such as an 'ease of doing business' index, a corruption index and figures on foreign direct investment. The proxy for market liquidity is the effective bid–ask spread computed using daily security prices (Roll, 1984). All specifications and sources of the control variables are summarized in Table A.4 of Appendix A.

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<sup>15</sup> This is in line with the literature. Karolyi and Stulz (1996) also looked at how correlations between stock markets change over time and found that they increase when market movement is high. Further, analyses performed across countries revealed that return linkages and volatility linkages increase in periods of stock market uncertainty, crisis periods or market crashes (Aggawal et al., 1999; Kallberg et al., 2005).

<sup>16</sup> Several empirical studies have included proxies for regulations when modelling volatility linkages, such as the deregulation following the October 1987 crash and the degree of bank intervention (Kanas, 2000), a regulation index (Bae et al., 2009), and the integration of financial markets (Kanas, 2000; Kim et al., 2001).

## 5. Empirical Results

### 5.1. Univariate Analysis

The univariate analysis is conducted to test whether there is a significant difference in volatility linkages when Islamic assets are involved. To this end, we consider the difference between the volatility correlations that involve two conventional assets and the volatility correlations that involve at least one Islamic asset. We estimate the volatility linkages across the whole sample period using the Pearson correlation (applied to either the absolute or the squared returns as volatility proxies), and the stochastic volatility model of Fleming et al. (1998) in a *GMM* setting. Based on the reasoning presented in previous sections, we expect the mean of this difference to be significant and positive. An excerpt of the volatility linkages estimated at the individual country level for a subset of countries and using all three methods is presented in Table B.2 of Appendix B.

#### 5.1.1 Volatility Linkages by Asset Type

Table 1 presents the results for all difference terms of volatility linkages in the far right column, as well as for subsets of difference terms by asset type.

**[Insert Table 1 about here]**

In Panels A and B, the test is conducted using the difference series of volatility linkages computed using the Pearson correlation. Panel A uses absolute returns as a proxy for volatility, while Panel B uses squared returns. Results are similar between the two panels, suggesting that, with the exception of two difference series containing only four values each, the differences are significantly different from zero at the 1% level, against both a two-sided and a one-sided alternative. On average, and using samples with at least 24 observations,



volatility linkages involving at least one Islamic asset are 2.13 to 3.01 percentage points lower than those between purely conventional assets. Using the whole sample of countries, they are 3.36 to 3.79 percentage points lower (and significant at the 1% level).

In Panel C, we conduct the test of significance in volatility correlations using the *GMM* method. The *GMM* results confirm that the sample means of the differences are mostly positive and even higher than those obtained using the Pearson correlation approach. However, the number of observations is lower and the differences at the asset type level are not always statistically significant.<sup>17</sup> Using samples with at least 10 observations, we find that volatility linkages involving at least one Islamic asset appear to be between 10.36 and 23.99 percentage points lower. However, using the whole sample, they are 13.53 percentage points lower (significant at the 1% level).

#### *5.1.2. Break Down by Country Type*

The main hypothesis that volatility linkages differ between Islamic and non-Islamic assets can be expanded to assess whether this difference is greater in Islamic countries. For example, Saiti et al. (2014) suggest that among a sample of nine emerging countries Islamic countries tend to provide better diversification benefits and are less volatile compared to non-Islamic countries. Table 2 extends the analysis from Table 1 by considering two samples: one with 9 Islamic countries, and one comprising 37 non-Islamic countries and the world index.

**[Insert Table 2 about here]**

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<sup>17</sup> We lose observations because the minimization does not always converge given that we have about 50 moment conditions and a relatively short period of 3 years.

First, we test for zero means within each sample, and then for equal means across samples. Results indicate that the sample mean of the difference in correlations between two conventional assets and those involving one Islamic asset is significant and positive in both Islamic and non-Islamic countries, with one exception in which we have only four observations. The test for equal means between the same differences gives mostly positive results, although these are not always significant. However, they are significant overall for correlations of absolute returns. Thus, the conclusion drawn in the previous section appears stronger in Islamic countries when using overall correlations. However, a more accurate picture is needed, and can be provided by multivariate analysis performed on time-varying monthly correlations.

## 5.2. Multivariate Analysis

The univariate analysis is now extended to a multivariate setup as described by Eqs. (22) and (23), where monthly volatility correlations represent the dependent variable, and the dummy variables denoting Islamic assets or countries and the various control variables are the explanatory regressors. The panel regression is run separately for the correlations between (i) stock and bond markets, and (ii) stock and money markets.<sup>18</sup> We assess the serial correlation in volatilities using the Durbin Watson test for panel regression with the critical values in Bhargava et al. (1982). All correlations appear serially correlated in the panel regressions;<sup>19</sup> however, one autocorrelation lag is enough to reject further serial correlation. All regressions are estimated using the *Newey West Heteroskedasticity Autocorrelation-Consistent (HAC)* standard errors. Since we need monthly values for volatility linkages, the volatility

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<sup>18</sup> The case for volatility correlations between bond and money markets cannot be tested, as there are no Islamic money assets in this study and only four Islamic bond assets (two of which do not even belong to individual countries, but rather to the GCC and world index).

<sup>19</sup> Results are not reported in the interest of brevity, but they are available upon request.

correlations are estimated at this frequency using the Pearson correlation (with either the absolute returns or the squared returns as the volatility proxy). The multivariate analysis is performed for the two cases mentioned above.

#### *5.2.1. Volatility Linkages between Islamic and Conventional Stock and Bond Markets*

Eq. (22) tests whether the volatility linkages between stocks and bonds differ between Islamic and conventional assets. The relevant coefficients for capturing this difference are: the Islamic country dummy variable ( $\beta_1$ ), which takes the value of 1 when volatility linkages are estimated in an Islamic country; the Islamic stock and Islamic bond dummy variable ( $\beta_2$ ), which takes the value of 1 when the volatility linkages are estimated between the Islamic stock and Islamic bond markets; the Islamic stock and conventional bond dummy variable ( $\beta_3$ ), which takes the value of 1 when the volatility linkages are estimated between the Islamic stock and conventional bond markets; and the Islamic bond and conventional stock dummy variable ( $\beta_4$ ), which takes the value of 1 when the volatility linkages are estimated between the conventional stock and Islamic bond markets. When volatility linkages are estimated between conventional stock and bond markets, the last three dummies are equal to 0. The listed coefficients are expected to be significantly negative if the volatility linkages that involve at least one Islamic asset are lower than those across conventional stock and bonds.

Table 3 presents the results for both the individual and the joint significance of the Islamic variables. It summarizes the tests conducted using various specifications: for the total sample in Panel A, and for two partitioned samples (Islamic and non-Islamic countries) in Panels B and C, using two different volatility proxies for the Pearson correlations, and with and without control variables and an autocorrelation lag correction term.

[Insert Table 3 about here]

In five (three) out of six specifications, the Islamic variables are jointly significant at the 10% (5%) level for the total sample.<sup>20</sup> Whenever the  $\widehat{\beta}_2$  coefficient estimate is individually significant, its value is negative, which means that volatility linkages between Islamic stock and bond indices are significantly lower than volatility linkages between conventional stock and bond indices (the reference pair of assets) by 6.75 to 8.50 percentage points. The  $\widehat{\beta}_3$  coefficient estimate is always negative and significant at the 5% level. Thus, volatility linkages between Islamic stock and conventional bond indices are lower than volatility linkages between conventional stock and bond indices by 1.91 to 2.28 percentage points. Finally, at the 5% level, the  $\widehat{\beta}_4$  coefficient estimate is either insignificant or significantly negative, indicating that volatility linkages between conventional stock and Islamic bond indices tend to be lower than those between conventional stock and bond indices by 4.69 to 6.45 percentage points. Overall, the regression from Eq. (22) is a good fit. The model specification with control variables and the  $AR(1)$  correction term achieves the highest explanatory power with an  $R_{adj}^2$  of 17.30% or 20.51%, depending on the volatility proxy. When only the control variables are included in the model, the  $R_{adj}^2$  values range between 14.08 and 15.44%, and the  $F$ -statistics are significant. However, when no control variables are included, the  $R_{adj}^2$  values fall to a range of 1.90 to 3.81% and the  $F$ -statistics become less significant. Thus, the control variables in Eq. (22) provide significant additional explanatory power, and we find most of them to be individually significant. In both markets, the volatility linkages are stronger when market frictions are low, the liquidity and volatility of the underlying assets are high, and during times of uncertainty and crisis.

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<sup>20</sup> We discuss the  $\widehat{\beta}_1$  estimate in Sub-Section 5.2.3.

When we break down the sample by country type, we find the  $\widehat{\beta}_3$  estimate to be reliably negative across samples, and larger and more significant in the Islamic sample. Specifically, in the Islamic country sample, six (four) specifications are significant at the 5% (1%) level, and in the non-Islamic countries sample, four (two) of six specifications are significant at the 10% (5%) level. This difference ranges between 1.35 and 1.71 percentage points in non-Islamic countries, and between 3.34 and 5.33 percentage points in Islamic countries.

Among the parameter estimates reported in Table 3,  $\widehat{\beta}_3$  is the most significant. Therefore, we next assess whether the Islamic asset effect (measured as the difference between volatility linkages across Islamic stock and conventional bond markets and volatility linkages across conventional stock and bond markets) is greater in Islamic countries than in non-Islamic ones. To this end, we compare the absolute values of the  $\widehat{\beta}_3$  coefficient estimates across the two samples. Results reported in the bottom row of Table 3 show that the difference between the absolute values of the  $\widehat{\beta}_3$  coefficients in the two samples is always positive, meaning that the Islamic asset effect as captured by  $\widehat{\beta}_3$  is greater in Islamic countries. The difference ranges between 1.63 and 3.98 percentage points.

### *5.2.2. Volatility Linkages between Islamic or Conventional Stocks and Conventional Money Markets*

Table 4 presents the results for the volatility linkages estimated between stock and money markets using the modified Eq. (23).

**[Insert Table 4 about here]**

The  $\widehat{\beta}_2$  coefficient estimate is significantly negative in six (three) out of six specifications at the 10% (5%) level in the total sample. As the estimated coefficient is always negative, volatility linkages between Islamic stock and conventional money market indices are lower than volatility linkages between conventional stock and money market indices by 2.40 to 2.62 percentage points in the total sample. Similarly, in the partitioned samples in Panels B and C, the  $\widehat{\beta}_2$  coefficient estimate is negative and of similar magnitude. However, it is significant in five out of six specifications in the non-Islamic sample. As above, the model with control variables and the  $AR(1)$  term achieves the highest explanatory power for the whole sample, as well as for the partitioned sub-samples, with a  $R_{adj}^2$  ranging from 20.87 to 38.17% across the two volatility proxies. In addition, the bottom row of Table 4 shows that the Islamic asset effect (measured as the difference between volatility linkages across Islamic stock and money markets and volatility linkages across conventional stock and money markets) is greater in Islamic countries than in non-Islamic ones. This difference ranges between 0.01 and 1.29 percentage points.

### 5.2.3. Comparison of Volatility Linkages in Islamic and Non-Islamic Countries

We also compare volatility linkages in Islamic versus non-Islamic countries by assessing the  $\widehat{\beta}_1$  (Islamic country dummy) coefficient estimate from Eqs. (22) and (23). Across all specifications in Tables 3 and 4, the Islamic country dummy is positive, suggesting that in general volatility linkages are higher in Islamic countries compared to non-Islamic ones. The coefficient estimate ranges between 5.86 and 13.57 percentage points and always has a statistically significant effect on the stock–bond markets correlation. For the stock–money markets correlation, its effect ranges between 3.59 and 6.79 percentage points (and is in most cases statistically significant). These results suggest that in general volatility linkages in

Islamic countries are stronger than in non-Islamic countries, holding all other factors constant. However, as highlighted in previous sections, correlations that involve at least one Islamic asset are lower than those between two conventional assets.

## 6. Robustness Analysis

### 6.1. Controlling for the Oil Effect

As highlighted by the results reported in Tables 3 and 4, the correlations that involve one Islamic asset (the Islamic stock) are lower in Islamic markets. It can be argued that Islamic equity indices are heavily dominated by the oil index and hence that our results may be driven by the oil sector rather than by the entire Islamic market (Ghorbel et al., 2014). Therefore, as a robustness check, in this section we purge the data of the oil index effect.

**[Insert Table 5 about here]**

To this end, we orthogonalise each country's equity index on the corresponding oil index. We then rerun all the tests performed in Section 5.1. Results are presented in Table 5, Panels A and B. They are robust and of similar magnitude to the results reported in Panels A and B of Table 1. This shows that our findings are not driven by the oil effect. In some instances, results are larger and significant when the oil index is removed from the equity indices. This suggests that our results are representative of the whole market and that they are not sector specific. Results are also robust when we perform the additional analysis of splitting the sample into Islamic and non-Islamic countries.<sup>21</sup>

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<sup>21</sup> Results are not reported here in the interest of brevity, but they are available upon request.

## 6.2. Crisis and Non-Crisis Sub-Periods

Various studies have found that volatility linkages increase in periods of stock market uncertainty, crisis periods or market crashes (Karolyi and Stulz, 1996; Aggarwal et al., 1999; Kallberg et al., 2005; Baruník et al., 2015). More importantly, higher volatility linkages have been observed during the recent past, due to the global financial crisis that started in 2008.<sup>22</sup> There is also some evidence that Islamic assets and financial institutions were affected by the crisis to a smaller degree than their conventional counterparts (Beck et al., 2013; Al-Khazali et al., 2014; Ho et al., 2014).<sup>23</sup> As a result, we divide our sample period into two distinct subsamples, covering non-crisis and crisis periods. While there are different opinions about the exact date of the onset of the US financial crisis, we use 17 March 2008 to 8 June 2010 as the crisis period. 17 March 2008 is the date on which US Investment Bank Bear Stearns & Co was taken over by JP Morgan (Manda, 2010), while 8 June 2010 is the latest observation in our sample, at which point the international crises were still ongoing.

**[Insert Table 6 about here]**

While our results remain robust across these two time periods, there are stronger diversification benefits from including at least one Islamic asset in the investment portfolio during the crisis period. As reflected by Table 6, the differences in volatility linkages are larger during the crisis period (see Panel A versus Panel C, or Panel B versus Panel D). The differences are significant at the 1% level overall, and also for pairs of assets with sample sizes above 24 observations.

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<sup>22</sup> For example, Aloui et al. (2015) find that the co-movement (measured as dynamic correlations) between Islamic stocks and bonds in the GCC increase when the global financial crisis spread globally in 2008-2009. Spillover effects are also generally larger during volatile (crisis) periods (Adams et al., 2014).

<sup>23</sup> See, for example Moody's Investor Service (2008), Abdul Aziz and Gintzburger (2009), Farooq (2009), Mirakhor and Krichene (2009), Akhtar (2010), Hasan and Dridi (2011), Smolo and Mirakhor (2010), Sukmana and Kholid (2010), Chapra (2011), Kaye and Hassan (2011), Dogarawa (2012), Mohieldin (2012), Beck et al. (2013), Al-Khazali et al. (2014), Ho et al. (2014), Jawadi et al. (2014), Rosman et al. (2014).



The next step is to assess the risk and return profiles of the Islamic and conventional indices. Table 7 presents annualized average returns and average standard deviations of returns for different asset classes, where the average is computed across countries, and for the world indices. We report results for the whole sample period, as well as for the non-crisis and crisis sub-periods. While equity (both conventional and Islamic) provided a higher return on average over our sample period, it has also been more volatile when compared to the other financial assets. This was found both overall and for the sub-periods under analysis (see Panels A, C and E). Results reported in Panels B, D and F for the world indices show that both the conventional and the Islamic world bond indices performed better in terms of returns over our sample period. Meanwhile, both the conventional and the Islamic equity world indices experienced lower (and negative) returns, with the Islamic one being consistently less affected across the different sample periods. Panels E and F show that the relative performance of Islamic equities has overall been marginally better than that of conventional equities during the crisis. However, they were not spared by the crisis and were affected through financial contagion resulting in lower returns and higher standard deviation relative to the pre-crisis period (see Yilmaz et al., 2015).

**[Insert Table 7 about here]**

### *6.3. Using an Alternative Volatility Model*

As an additional robustness check, we consider *GARCH*-implied volatilities in our analysis. To this end, we perform the *GARCH* estimation for all individual return series. Based on the Akaike and the Schwarz Bayesian information criteria, we choose a *GARCH(1,1)* model specification for our various return series, with series-specific parameter estimates.

Robustness checks are then performed by computing correlations between *GARCH*-implied volatility series for the same pairs as in Tables 1 and 2.

**[Insert Table 8 about here]**

As can be seen in Table 8, results are robust. Although they are not as significant as the results reported in Table 1 (we are working with implied volatilities now, not realized ones), they are in line with them. These results imply that the linkages between implied volatilities are lower when an Islamic asset is included in the comparison. As before, when we break down our sample by country type, the differences remain larger in Islamic countries, versus non-Islamic ones (see Panel B in Table 8).<sup>24</sup>

## **7. Conclusion**

Despite the repeated injection of billions of dollars into markets by central banks on both sides of the Atlantic, the recent financial crises have seen stock markets experience unprecedented volatility. Most asset classes have experienced significant pullbacks, with strong cross-correlations across asset classes. The channels of transmission of

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<sup>24</sup> We perform some additional robustness checks for sub-samples of countries for which we have the available data: (i) we extended the time period back to 31 May 2005 for Canada, the GCC, Malaysia, the UAE, the UK and the US, and back to 31 December 1999 for Canada, the UK and the US; (ii) we used alternative bond and money market indices for Australia, Hungary, Indonesia, Malaysia, Pakistan and South Korea; and (iii) we used alternative stock indices, where the standard stock indices (which consist of mid- and large-cap firms) were substituted with large-cap, mid-cap, small-cap, SMID (small- and mid-cap) and IMI (small-, mid- and large-cap) stock indices for Australia, Canada, Egypt, Hungary, Indonesia, Malaysia, Pakistan, South Korea, the UK and the US. The tests performed in (ii) and (iii) verified that the difference in methodologies across index providers did not affect our results. Results are documented in Tables B.3–B.5 of Appendix B and support robustness. Details on the alternative indices are provided in Panel B of Table A.2 and in Panel B of Table A.3 of Appendix A.

macroeconomic and financial risks resided in the global diversification of financial portfolios. This phenomenon translated into the rapid spread of volatility through global financial markets. However, looking at Islamic assets, the financial contagion seems to have affected them to a lesser degree.

While volatility linkages and Islamic finance have been researched individually, we take the first step to combine these fields and examine the extent to which volatility linkages differ when Islamic assets are involved. More specifically, we examine how volatility linkages across stock, bond and money market indices differ when Islamic or conventional assets are traded in Islamic and non-Islamic countries. While we find volatility linkages to vary significantly across countries, there is also a specific pattern across Islamic assets.

First, volatility linkages are weaker in Islamic markets relative to non-Islamic markets, as the former are characterized by a smaller set of common information and lower cross-market hedging activity. Overall, the difference between volatility linkages that involve at least one Islamic asset and volatility linkages between two conventional assets ranges from 3.36 to 13.53 percentage points.

Second, the difference in volatility linkages between Islamic and non-Islamic assets is greater in Islamic countries when Islamic stocks are included in the comparison. While this difference ranges between 0.70 and 7.07 percentage points, it remains statistically significant across various specifications. We attribute this effect to the Islamic principles and regulations that only apply in Islamic countries. The weaker effect in non-Islamic countries appears to be due to Islamic stock indices excluding certain industries or firms that do not comply with Islamic law.

Third, we show that our results are not driven by the oil sector. They also hold during both crisis and non-crisis periods, and are stronger during the recent global financial crisis.

Our findings have important implications for investment and risk management, especially in view of the increasing popularity of Islamic finance both within and beyond Islamic countries. Our results suggest that Islamic assets may have been less contaminated by the recent financial crisis. This is an area of increasing interest, and as new countries open up to Islamic finance, it will be interesting to observe how volatility linkages involving Islamic assets evolve over time.

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**Table 1: Differences between Volatility Linkages that Involve Two Conventional Assets, and Volatility Linkages that Involve at Least One Islamic Asset**

Table 1 presents the results of testing for zero mean using five difference series of daily volatility linkages and all the difference series combined. The difference series of volatility linkages are obtained by subtracting the correlation that involves at least one Islamic (IS) asset from the corresponding correlation between conventional (C) assets. Assets include stock, government (Govt) bond, corporate (Corp) bond and money instruments. Data cover the period May 2007 to June 2010. The correlations are estimated using Pearson correlation across absolute returns (Panel A) and squared returns (Panel B), as well as *GMM* estimates based on Fleming, Kirby, and Ostdiek's (1998) stochastic volatility model and the arising moment conditions from Eq. (20) (Panel C). The lag length in all *GMM* models is  $L = 40$ . Estimates that are significant at the 1%, 5% and 10% levels are denoted by superscripts a, b and c, respectively. Next to the t-statistics, these superscripts denote one-tailed tests, and next to the p-values, they refer to two-tailed tests.

| <i>Differences</i>                               | <i>C Stock and C Govt Bond - IS Stock and C Govt Bond</i> | <i>C Stock and C Corp Bond - IS Stock and C Corp Bond</i> | <i>C Stock and C Money - IS Stock and C Money</i> | <i>C Stock and C Corp Bond - C Stock and IS Corp Bond</i> | <i>C Stock and C Corp Bond - IS Stock and IS Corp Bond</i> | <i>Total Differences</i> |
|--------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|--------------------------|
| <i>Panel A: Correlations of Absolute Returns</i> |                                                           |                                                           |                                                   |                                                           |                                                            |                          |
| Observations                                     | 37                                                        | 27                                                        | 24                                                | 4                                                         | 4                                                          | 96                       |
| Mean                                             | 0.0285                                                    | 0.0260                                                    | 0.0287                                            | 0.0955                                                    | 0.2019                                                     | 0.0379                   |
| T-Statistic                                      | 4.3274 <sup>a</sup>                                       | 4.9046 <sup>a</sup>                                       | 4.3816 <sup>a</sup>                               | 1.3544                                                    | 2.4407 <sup>b</sup>                                        | 5.9217 <sup>a</sup>      |
| P-value                                          | 0.0001 <sup>a</sup>                                       | 0.0000 <sup>a</sup>                                       | 0.0002 <sup>a</sup>                               | 0.2686                                                    | 0.0924 <sup>c</sup>                                        | 0.0000 <sup>a</sup>      |
| <i>Panel B: Correlations of Squared Returns</i>  |                                                           |                                                           |                                                   |                                                           |                                                            |                          |
| Observations                                     | 37                                                        | 27                                                        | 24                                                | 4                                                         | 4                                                          | 96                       |
| Mean                                             | 0.0301                                                    | 0.0252                                                    | 0.0213                                            | 0.1030                                                    | 0.1272                                                     | 0.0336                   |
| T-Statistic                                      | 3.0203 <sup>a</sup>                                       | 3.5738 <sup>a</sup>                                       | 3.2484 <sup>a</sup>                               | 1.3609                                                    | 1.7725 <sup>c</sup>                                        | 5.1951 <sup>a</sup>      |
| P-value                                          | 0.0046 <sup>a</sup>                                       | 0.0014 <sup>a</sup>                                       | 0.0035 <sup>a</sup>                               | 0.2668                                                    | 0.1744                                                     | 0.0000 <sup>a</sup>      |
| <i>Panel C: GMM Estimates</i>                    |                                                           |                                                           |                                                   |                                                           |                                                            |                          |
| Observations                                     | 10                                                        | 12                                                        | 10                                                | 4                                                         | 3                                                          | 39                       |
| Mean                                             | 0.1036                                                    | 0.2399                                                    | 0.1132                                            | -0.0075                                                   | 0.8677                                                     | 0.1353                   |
| T-Statistic                                      | 1.0155                                                    | 2.1822 <sup>b</sup>                                       | 1.8632 <sup>b</sup>                               | -0.2974                                                   | 1.4657                                                     | 2.9455 <sup>a</sup>      |
| P-value                                          | 0.3364                                                    | 0.0517 <sup>c</sup>                                       | 0.0953 <sup>c</sup>                               | 0.7856                                                    | 0.2804                                                     | 0.0055 <sup>a</sup>      |

**Table 2: Differences by Country Type**

Table 2 presents the results of testing for zero mean using three difference series of volatility linkages and a combination of them. The difference series are obtained by subtracting the correlation that involves at least one Islamic (IS) asset from the corresponding correlation across two conventional (C) assets. Assets include stock, government (Govt) bond, corporate (Corp) bond and money instruments. The correlations are estimated using Pearson correlation across absolute returns (Panel A) and squared returns (Panel B).<sup>25</sup> Data cover the period May 2007 to June 2010. The total sample is partitioned into Islamic countries and non-Islamic countries. In each panel, the 1%, 5% and 10% levels of significance are denoted by superscripts a, b and c, respectively. Next to the t-statistics, these superscripts denote one-tailed tests, and next to the p-values, they refer to two-tailed tests.

| <i>Differences</i>                               | <i>C Stock &amp; C Govt Bond - IS Stock &amp; C Govt Bond</i> |                     | <i>C Stock &amp; C Corp Bond - IS Stock &amp; C Corp Bond</i> |                     | <i>C Stock &amp; C Money - IS Stock &amp; C Money</i> |                     | <i>Total Differences</i> |                     |
|--------------------------------------------------|---------------------------------------------------------------|---------------------|---------------------------------------------------------------|---------------------|-------------------------------------------------------|---------------------|--------------------------|---------------------|
| Sample                                           | Islamic                                                       | Non-Islamic         | Islamic                                                       | Non-Islamic         | Islamic                                               | Non-Islamic         | Islamic                  | Non-Islamic         |
| Observations                                     | 3                                                             | 34                  | 9                                                             | 18                  | 4                                                     | 20                  | 16                       | 72                  |
| <i>Panel A: Correlations of Absolute Returns</i> |                                                               |                     |                                                               |                     |                                                       |                     |                          |                     |
| Mean                                             | 0.0707                                                        | 0.0248              | 0.0371                                                        | 0.0205              | 0.0350                                                | 0.0274              | 0.0429                   | 0.0244              |
| T-Statistic                                      | 3.8728 <sup>b</sup>                                           | 3.7129 <sup>a</sup> | 3.2704 <sup>a</sup>                                           | 3.8097 <sup>a</sup> | 1.4252                                                | 4.1969 <sup>a</sup> | 4.5434 <sup>a</sup>      | 6.3593 <sup>a</sup> |
| P-value                                          | 0.0607 <sup>c</sup>                                           | 0.0008 <sup>a</sup> | 0.0113 <sup>b</sup>                                           | 0.0014 <sup>a</sup> | 0.2493                                                | 0.0005 <sup>a</sup> | 0.0004 <sup>a</sup>      | 0.0000 <sup>a</sup> |
| <i>Test for Equal Means</i>                      |                                                               |                     |                                                               |                     |                                                       |                     |                          |                     |
| Difference                                       | 0.0459                                                        |                     | 0.0166                                                        |                     | 0.0076                                                |                     | 0.0185                   |                     |
| T-Statistic                                      | 1.9786 <sup>b</sup>                                           |                     | 1.5098 <sup>c</sup>                                           |                     | 0.4257                                                |                     | 1.9882 <sup>b</sup>      |                     |
| P-value                                          | 0.0558 <sup>c</sup>                                           |                     | 0.1436                                                        |                     | 0.6744                                                |                     | 0.0500 <sup>b</sup>      |                     |
| <i>Panel B: Correlations of Squared Returns</i>  |                                                               |                     |                                                               |                     |                                                       |                     |                          |                     |
| Mean                                             | 0.0607                                                        | 0.0274              | 0.0309                                                        | 0.0224              | 0.0070                                                | 0.0242              | 0.0305                   | 0.0253              |
| T-Statistic                                      | 4.0428 <sup>b</sup>                                           | 2.5666 <sup>a</sup> | 2.3689 <sup>b</sup>                                           | 2.6233 <sup>a</sup> | 0.4045                                                | 3.4010 <sup>a</sup> | 3.1776 <sup>a</sup>      | 4.3832 <sup>a</sup> |
| P-value                                          | 0.0561 <sup>c</sup>                                           | 0.0150 <sup>b</sup> | 0.0453 <sup>b</sup>                                           | 0.0178 <sup>b</sup> | 0.7130                                                | 0.0030 <sup>a</sup> | 0.0062 <sup>a</sup>      | 0.0000 <sup>a</sup> |
| <i>Test for Equal Means</i>                      |                                                               |                     |                                                               |                     |                                                       |                     |                          |                     |
| Difference                                       | 0.0333                                                        |                     | 0.0085                                                        |                     | -0.0172                                               |                     | 0.0052                   |                     |
| T-Statistic                                      | 0.9094                                                        |                     | 0.5556                                                        |                     | 0.9765                                                |                     | 0.4001                   |                     |
| P-value                                          | 0.3694                                                        |                     | 0.5834                                                        |                     | 0.3395                                                |                     | 0.6901                   |                     |

<sup>25</sup> The analysis was also conducted using differences between volatility linkages that were estimated using the stochastic volatility model in *GMM*. However, except for the series that includes all differences, the number of observations was only between 1 and 6, and any inferences from tests for equal means between the series were limited. The sample means are higher (between 0.1077 and 0.3497) than are those in Panels A and B, but they are only statistically significant when the number of observations is 6 or above.

**Table 3: Multivariate Analysis of the Islamic Assets Effect on the Stock-Bond Volatility Linkages**

Table 3 presents individual coefficient estimates with their t-statistics for all Islamic asset variables, as well as a joint test of these variables. The volatility correlations are estimated between stock and bond markets. Different versions of the model below are tested: without control variables, with control variables, and with control variables and an  $AR(1)$  term:

$$\rho_{i,a,b,t} = \beta_0 + \beta_1 I_{\{\text{Islamic Country}\}} + \beta_2 I_{\{\text{Islamic Stock and Islamic Bond}\}} + \beta_3 I_{\{\text{Islamic Stock and Conventional Bond}\}} + \beta_4 I_{\{\text{Islamic Bond and Conventional Stock}\}} + \sum_{m=6}^n \beta_{i,m} x_{i,m} + \varepsilon_{i,t}$$

The control group for dummies 2–4 is when the volatility linkage is estimated between conventional stock and bond. Panels A, B and C refer to the total, Islamic countries and non-Islamic countries samples, respectively. Each specification is run using either absolute returns or squared returns as the volatility proxy. The Islamic country variable is only meaningful in the total sample, and is excluded from panels B and C. Data cover the period May 2007 to June 2010. Superscripts a, b and c denote the 1%, 5% and 10% levels of significance, respectively. Superscripts next to the t-statistics and p-values refer to one-tailed tests and two-tailed tests, respectively. The  $R^2_{\text{adj}}$  is reported in percentages.

| Test                                                                | No control variables |                      | All control variables |                      | All control variables and $AR(1)$ |                      |
|---------------------------------------------------------------------|----------------------|----------------------|-----------------------|----------------------|-----------------------------------|----------------------|
| Volatility Proxy                                                    | Abs Return           | Sqrd Return          | Abs Return            | Sqrd Return          | Abs Return                        | Sqrd Return          |
| <i>Panel A: Total Sample (Observations=4773)</i>                    |                      |                      |                       |                      |                                   |                      |
| $\hat{\beta}_1$                                                     | 0.0586               | 0.0803               | 0.0887                | 0.1357               | 0.0822                            | 0.1249               |
| T-Statistic                                                         | 5.3070 <sup>a</sup>  | 6.2006 <sup>a</sup>  | 5.7948 <sup>a</sup>   | 8.0568 <sup>a</sup>  | 4.2973 <sup>a</sup>               | 5.6248 <sup>a</sup>  |
| $\hat{\beta}_2$                                                     | -0.0107              | -0.0850              | 0.0298                | -0.0675              | 0.0263                            | -0.0717              |
| T-Statistic                                                         | -0.3990              | -2.992 <sup>c</sup>  | 1.1763                | -2.4921 <sup>a</sup> | 0.8118                            | -1.9826 <sup>a</sup> |
| $\hat{\beta}_3$                                                     | -0.0203              | -0.0213              | -0.0196               | -0.0228              | -0.0191                           | -0.0212              |
| T-Statistic                                                         | -2.3911 <sup>a</sup> | -2.1500 <sup>a</sup> | -2.3969 <sup>a</sup>  | -2.4443 <sup>a</sup> | -1.8471 <sup>b</sup>              | -1.7121 <sup>b</sup> |
| $\hat{\beta}_4$                                                     | -0.0025              | -0.0645              | 0.0380                | -0.0469              | 0.0348                            | -0.0495              |
| T-Statistic                                                         | -0.0937              | -2.2573 <sup>a</sup> | 1.4935 <sup>c</sup>   | -1.7428 <sup>b</sup> | 1.0679                            | -1.3591 <sup>c</sup> |
| $R^2_{\text{adj}}$                                                  | 3.81                 | 1.90                 | 14.08                 | 15.44                | 17.30                             | 20.51                |
| <i>Joint Test</i>                                                   |                      |                      |                       |                      |                                   |                      |
| F-Statistic                                                         | 1.9142               | 2.2926               | 3.7968                | 3.7329               | 2.1180                            | 2.1451               |
| P-value                                                             | 0.1250               | 0.0760 <sup>b</sup>  | 0.0098 <sup>a</sup>   | 0.0108 <sup>b</sup>  | 0.0958 <sup>c</sup>               | 0.0924 <sup>c</sup>  |
| <i>Panel B: Islamic countries (Observations=1100)</i>               |                      |                      |                       |                      |                                   |                      |
| $\hat{\beta}_2$                                                     | 0.0441               | -0.0625              | 0.0110                | -0.0162              | 0.0037                            | -0.0219              |
| T-Statistic                                                         | 1.3671               | -1.7556 <sup>b</sup> | 0.3590                | -0.4883              | 0.1089                            | -0.5859              |
| $\hat{\beta}_3$                                                     | -0.0334              | -0.0533              | -0.0337               | -0.0533              | -0.0345                           | -0.0526              |
| T-Statistic                                                         | -1.8027 <sup>b</sup> | -2.3368 <sup>a</sup> | -2.0137 <sup>a</sup>  | -2.9149 <sup>a</sup> | -1.8550 <sup>b</sup>              | -2.5142 <sup>a</sup> |
| $\hat{\beta}_4$                                                     | 0.0550               | -0.0336              | 0.0220                | 0.0127               | 0.0147                            | 0.0087               |
| T-Statistic                                                         | 1.6469               | -0.9265              | 0.6991                | 0.3764               | 0.4230                            | 0.2273               |
| $R^2_{\text{adj}}$                                                  | 17.54                | 9.72                 | 34.40                 | 39.90                | 34.28                             | 41.16                |
| <i>Joint Test</i>                                                   |                      |                      |                       |                      |                                   |                      |
| F-Statistic                                                         | 4.0880               | 2.2167               | 2.0386                | 3.3125               | 1.4955                            | 2.3982               |
| P-value                                                             | 0.0067 <sup>a</sup>  | 0.0846 <sup>c</sup>  | 0.1068                | 0.0195 <sup>b</sup>  | 0.2142                            | 0.0666 <sup>c</sup>  |
| <i>Panel C: Non-Islamic countries (Observations=3663)</i>           |                      |                      |                       |                      |                                   |                      |
| $\hat{\beta}_2$                                                     | -0.0549              | -0.0838              | -0.0371               | -0.0645              | -0.0247                           | -0.0500              |
| T-Statistic                                                         | -1.0358              | -1.5637 <sup>c</sup> | -0.6926               | -1.2004              | -0.3752                           | -0.7282              |
| $\hat{\beta}_3$                                                     | -0.0171              | -0.0135              | -0.0164               | -0.0164              | -0.0154                           | -0.0146              |
| T-Statistic                                                         | -1.8197 <sup>a</sup> | -1.2348              | -1.8374 <sup>b</sup>  | -1.6091 <sup>c</sup> | -1.4048 <sup>c</sup>              | -1.1238              |
| $\hat{\beta}_4$                                                     | -0.0548              | -0.0888              | -0.0370               | -0.0695              | -0.0254                           | -0.0566              |
| T-Statistic                                                         | -1.0990              | -1.7120 <sup>b</sup> | -0.7555               | -1.3869              | -0.4185                           | -0.8640              |
| $R^2_{\text{adj}}$                                                  | 1.90                 | 0.61                 | 12.54                 | 13.79                | 14.93                             | 17.47                |
| <i>Joint Test</i>                                                   |                      |                      |                       |                      |                                   |                      |
| F-Statistic                                                         | 1.7358               | 1.9007               | 1.3968                | 1.7137               | 0.7214                            | 0.7444               |
| P-value                                                             | 0.1574               | 0.1272               | 0.2418                | 0.1620               | 0.5391                            | 0.5255               |
| $ \hat{\beta}_3 (\text{Panel B}) -  \hat{\beta}_3 (\text{Panel C})$ | 0.0163               | 0.0398               | 0.0173                | 0.0368               | 0.0191                            | 0.0380               |

**Table 4: Multivariate Analysis of the Islamic Asset Effect on the Stock-Money Market Volatility Linkages**

Table 4 reports the results for the individual significance of the Islamic stock and conventional money market variable (coefficient  $\beta_2$  below). The model is run with and without control variables and an  $AR(1)$  term:

$$\rho_{i,a,b,t} = \beta_0 + \beta_1 I_{\{Islamic\ Country\}} + \beta_2 I_{\{Islamic\ Stock\ and\ Money\ Market\}} + \sum_{m=6}^n \beta_{i,m} x_{i,m} + \varepsilon_{i,t}$$

The control group for dummy 2 is when the volatility linkage is estimated between conventional stock and money market. Panels A, B and C include the  $\beta_2$  coefficient estimate and the corresponding t-statistic and p-value for the total sample, sample A with only Islamic countries and sample B with only non-Islamic countries, respectively. Each specification is run using either absolute returns or squared returns as the volatility proxy. Data cover the period May 2007 to June 2010. Superscripts a, b and c denote the 1%, 5% and 10% levels of significance, respectively. Superscripts next to the t-statistics and p-values refer to one-tailed tests and two-tailed tests, respectively. The  $R^2_{adj}$  is reported in percentages.

| Test                                                                   | No control variables |                      | All control variables |                      | All control variables and $AR(1)$ |                      |
|------------------------------------------------------------------------|----------------------|----------------------|-----------------------|----------------------|-----------------------------------|----------------------|
| Volatility Proxy                                                       | Abs Return           | Sqrd Return          | Abs Return            | Sqrd Return          | Abs Return                        | Sqrd Return          |
| <i>Panel A: Total Sample (Observations = 1739)</i>                     |                      |                      |                       |                      |                                   |                      |
| $\widehat{\beta}_1$                                                    | 0.0359               | 0.0371               | 0.0513                | 0.0679               | 0.0463                            | 0.0574               |
| T-Statistic                                                            | 1.7502 <sup>b</sup>  | 1.6673 <sup>b</sup>  | 2.0318 <sup>a</sup>   | 2.4790 <sup>a</sup>  | 1.4393                            | 1.7419 <sup>b</sup>  |
| $\widehat{\beta}_2$                                                    | -0.0240              | -0.0239              | -0.0262               | -0.0259              | -0.0255                           | -0.0251              |
| T-Statistic                                                            | -1.6775 <sup>b</sup> | -1.4956 <sup>c</sup> | -2.0525 <sup>a</sup>  | -1.7843 <sup>b</sup> | -1.5218 <sup>c</sup>              | -1.4076 <sup>c</sup> |
| P-value                                                                | 0.0936 <sup>c</sup>  | 0.1349               | 0.0403 <sup>b</sup>   | 0.0745 <sup>c</sup>  | 0.1283                            | 0.1594               |
| $R^2_{adj}$                                                            | 0.95                 | 0.66                 | 21.81                 | 18.89                | 25.83                             | 20.87                |
| <i>Panel B: Sample A (Islamic countries) (Observations = 296)</i>      |                      |                      |                       |                      |                                   |                      |
| $\widehat{\beta}_2$                                                    | -0.0299              | -0.0346              | -0.0304               | -0.0355              | -0.0272                           | -0.0297              |
| T-Statistic                                                            | -0.8058              | -0.8668              | -1.0446               | -1.0985              | -0.7766                           | -0.8332              |
| P-value                                                                | 0.4210               | 0.3868               | 0.2971                | 0.2729               | 0.4381                            | 0.4055               |
| $R^2_{adj}$                                                            | -0.12                | -0.08                | 38.28                 | 34.50                | 38.17                             | 33.44                |
| <i>Panel C: Sample B (Non-Islamic countries) (Observations = 1446)</i> |                      |                      |                       |                      |                                   |                      |
| $\widehat{\beta}_2$                                                    | -0.0227              | -0.0217              | -0.0274               | -0.0264              | -0.0271                           | -0.0265              |
| T-Statistic                                                            | -1.4723 <sup>c</sup> | -1.2448              | -2.0121 <sup>b</sup>  | -1.6779 <sup>b</sup> | -1.6089 <sup>c</sup>              | -1.4267 <sup>c</sup> |
| P-value                                                                | 0.1411               | 0.2134               | 0.0444 <sup>b</sup>   | 0.0936 <sup>c</sup>  | 0.1079                            | 0.1539               |
| $R^2_{adj}$                                                            | 1.13                 | 0.76                 | 24.47                 | 20.81                | 26.55                             | 21.67                |
| $ \widehat{\beta}_2 (Panel\ B) -  \widehat{\beta}_2 (Panel\ C)$        | 0.0072               | 0.0129               | 0.0030                | 0.0091               | 0.0001                            | 0.0033               |

**Table 5: Differences between Volatility Linkages that Involve Two Conventional Assets, and Volatility Linkages that Involve at Least One Islamic Asset (Controlling for the Oil Index)**

Table 5 reports the results of testing for zero mean using five difference series of daily volatility linkages and all the combined difference series, after orthogonalising them on the oil index. The difference series of volatility linkages are obtained by subtracting the correlation that involves at least one Islamic (IS) asset from the corresponding correlation between conventional (C) assets. Assets include stock, government (Govt) bond, corporate (Corp) bond and money instruments. The correlations are estimated using Pearson correlation across absolute returns (Panel A) and squared returns (Panel B). Data cover the period May 2007 to June 2010. Estimates that are significant at the 1%, 5% and 10% levels are denoted by superscripts a, b and c, respectively. Next to the t-statistics, these superscripts denote one-tailed tests, and next to the p-values, they refer to two-tailed tests.

| <i>Differences</i>                               | <i>C Stock and C<br/>Govt Bond - IS<br/>Stock and C Govt<br/>Bond</i> | <i>C Stock and C<br/>Corp Bond - IS<br/>Stock and C Corp<br/>Bond</i> | <i>C Stock and C<br/>Money - IS Stock<br/>and C Money</i> | <i>C Stock and C<br/>Corp Bond - C<br/>Stock and IS<br/>Corp Bond</i> | <i>C Stock and C<br/>Corp Bond - IS<br/>Stock and IS<br/>Corp Bond</i> | <i>Total Differences</i> |
|--------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------|
| <i>Panel A: Correlations of Absolute Returns</i> |                                                                       |                                                                       |                                                           |                                                                       |                                                                        |                          |
| Observations                                     | 37                                                                    | 27                                                                    | 24                                                        | 4                                                                     | 4                                                                      | 96                       |
| Mean                                             | 0.0262                                                                | 0.0260                                                                | 0.0267                                                    | 0.0801                                                                | 0.1038                                                                 | 0.0316                   |
| T-Statistic                                      | 3.747 <sup>a</sup>                                                    | 4.869 <sup>a</sup>                                                    | 4.1244 <sup>a</sup>                                       | 1.3183                                                                | 2.4407 <sup>b</sup>                                                    | 6.5282 <sup>a</sup>      |
| P-value                                          | 0.0006 <sup>a</sup>                                                   | 0.0000 <sup>a</sup>                                                   | 0.0004 <sup>a</sup>                                       | 0.2790                                                                | 0.1001 <sup>c</sup>                                                    | 0.0000 <sup>a</sup>      |
| <i>Panel B: Correlations of Squared Returns</i>  |                                                                       |                                                                       |                                                           |                                                                       |                                                                        |                          |
| Observations                                     | 37                                                                    | 27                                                                    | 24                                                        | 4                                                                     | 4                                                                      | 96                       |
| Mean                                             | 0.0302                                                                | 0.0282                                                                | 0.0200                                                    | 0.1110                                                                | 0.1351                                                                 | 0.0348                   |
| T-Statistic                                      | 2.7092 <sup>a</sup>                                                   | 4.1723 <sup>a</sup>                                                   | 3.2001 <sup>a</sup>                                       | 1.7538 <sup>c</sup>                                                   | 2.3248 <sup>c</sup>                                                    | 5.4052 <sup>a</sup>      |
| P-value                                          | 0.0103 <sup>a</sup>                                                   | 0.0003 <sup>a</sup>                                                   | 0.0040 <sup>a</sup>                                       | 0.1777                                                                | 0.1026                                                                 | 0.0000 <sup>a</sup>      |

**Table 6: Differences between Volatility Linkages that Involve Two Conventional Assets, and Volatility Linkages that Involve at Least One Islamic Asset during Crisis and Non-Crisis Sub-Periods**

Table 6 reports the results of testing for zero mean using five difference series of daily volatility linkages and all the combined difference series, for non-crisis and crisis sub-periods. The difference series of volatility linkages are obtained by subtracting the correlation that involves at least one Islamic (IS) asset from the corresponding correlation between conventional (C) assets. Assets include stock, government (Govt) bond, corporate (Corp) bond and money instruments. The correlations are estimated using Pearson correlation across absolute returns (Panel A) and squared returns (Panel B). The entire sample period is 3 May 2007 to 8 June 2010 and is divided into a non-crisis sub-period (31 May 2007 to 16 March 2008) and a crisis sub-period (17 March 2008 to 8 June 2010). Estimates that are significant at the 1%, 5% and 10% levels are denoted by superscripts a, b and c, respectively. Next to the t-statistics, these superscripts denote one-tailed tests, and next to the p-values, they refer to two-tailed tests.

| <i>Differences</i>                                                                                        | <i>C Stock and C<br/>Govt Bond - IS<br/>Stock and C Govt<br/>Bond</i> | <i>C Stock and C<br/>Corp Bond - IS<br/>Stock and C Corp<br/>Bond</i> | <i>C Stock and C<br/>Money - IS Stock<br/>and C Money</i> | <i>C Stock and C<br/>Corp Bond - C<br/>Stock and IS<br/>Corp Bond</i> | <i>C Stock and C<br/>Corp Bond - IS<br/>Stock and IS<br/>Corp Bond</i> | <i>Total Differences</i> |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------|
| <i>Panel A: Correlations of Absolute Returns during the Non-crisis period (31 May 2007–16 March 2008)</i> |                                                                       |                                                                       |                                                           |                                                                       |                                                                        |                          |
| Observations                                                                                              | 37                                                                    | 27                                                                    | 24                                                        | 4                                                                     | 4                                                                      | 96                       |
| Mean                                                                                                      | 0.0246                                                                | 0.0228                                                                | 0.0322                                                    | 0.0412                                                                | 0.0690                                                                 | 0.0284                   |
| T-Statistic                                                                                               | 3.5932 <sup>a</sup>                                                   | 2.7950 <sup>a</sup>                                                   | 3.4548 <sup>a</sup>                                       | 1.5570                                                                | 2.2519 <sup>c</sup>                                                    | 6.0807 <sup>a</sup>      |
| P-value                                                                                                   | 0.0013 <sup>a</sup>                                                   | 0.0082 <sup>a</sup>                                                   | 0.0022 <sup>a</sup>                                       | 0.2173                                                                | 0.1097                                                                 | 0.0000 <sup>a</sup>      |
| <i>Panel B: Correlations of Squared Returns during the Non-crisis period (31 May 2007–16 March 2008)</i>  |                                                                       |                                                                       |                                                           |                                                                       |                                                                        |                          |
| Observations                                                                                              | 37                                                                    | 27                                                                    | 24                                                        | 4                                                                     | 4                                                                      | 96                       |
| Mean                                                                                                      | 0.0270                                                                | 0.0232                                                                | 0.0410                                                    | 0.0483                                                                | 0.0688                                                                 | 0.0321                   |
| T-Statistic                                                                                               | 2.9748 <sup>a</sup>                                                   | 2.8136 <sup>a</sup>                                                   | 3.4864 <sup>a</sup>                                       | 1.0644                                                                | 1.3381 <sup>b</sup>                                                    | 5.5742 <sup>a</sup>      |
| P-value                                                                                                   | 0.0052 <sup>a</sup>                                                   | 0.0092 <sup>a</sup>                                                   | 0.0020 <sup>a</sup>                                       | 0.3652                                                                | 0.2732 <sup>c</sup>                                                    | 0.0000 <sup>a</sup>      |
| <i>Panel C: Correlations of Absolute Returns during the Crisis period (17 March 2008–8 June 2010)</i>     |                                                                       |                                                                       |                                                           |                                                                       |                                                                        |                          |
| Observations                                                                                              | 37                                                                    | 27                                                                    | 24                                                        | 4                                                                     | 4                                                                      | 96                       |
| Mean                                                                                                      | 0.0315                                                                | 0.0252                                                                | 0.0363                                                    | 0.0713                                                                | 0.0787                                                                 | 0.0346                   |
| T-Statistic                                                                                               | 4.6763 <sup>a</sup>                                                   | 4.9047 <sup>a</sup>                                                   | 4.1738 <sup>a</sup>                                       | 1.4160                                                                | 1.5514                                                                 | 5.0470 <sup>a</sup>      |
| P-value                                                                                                   | 0.0000 <sup>a</sup>                                                   | 0.0000 <sup>a</sup>                                                   | 0.0004 <sup>a</sup>                                       | 0.2518                                                                | 0.2186                                                                 | 0.0000 <sup>a</sup>      |
| <i>Panel D: Correlations of Squared Returns during the Crisis period (17 March 2008–8 June 2010)</i>      |                                                                       |                                                                       |                                                           |                                                                       |                                                                        |                          |
| Observations                                                                                              | 37                                                                    | 27                                                                    | 24                                                        | 4                                                                     | 4                                                                      | 96                       |
| Mean                                                                                                      | 0.0345                                                                | 0.0289                                                                | 0.0365                                                    | 0.0642                                                                | 0.0704                                                                 | 0.0361                   |
| T-Statistic                                                                                               | 3.2264 <sup>a</sup>                                                   | 3.7117 <sup>a</sup>                                                   | 3.8140 <sup>a</sup>                                       | 0.9142                                                                | 1.0226                                                                 | 5.6707 <sup>a</sup>      |
| P-value                                                                                                   | 0.0027 <sup>a</sup>                                                   | 0.0010 <sup>a</sup>                                                   | 0.0009 <sup>a</sup>                                       | 0.9143                                                                | 0.3818 <sup>c</sup>                                                    | 0.0000 <sup>a</sup>      |

**Table 7: Average Risk-Return Profiles for Conventional and Islamic Asset Returns**

This table reports the annualized mean returns and standard deviations (using 252 trading days per year) of daily returns for conventional stocks, corporate bonds, government (Govt) bonds and money, and Islamic stocks and bonds. We report average results across all countries, and results for world indices. Panels A–B provide results for the entire period, while Panels C–D and E–F focus on the non-crisis and crisis sub-periods, respectively. The entire sample period is 31 May 2007 to 8 June 2010 and is divided into a non-crisis sub-period (31 May 2007 to 16 March 2008) and a crisis sub-period (17 March 2008 to 8 June 2010).

|                                                                               | <i>Conventional<br/>Stock</i> | <i>Islamic<br/>Stock</i> | <i>Money</i> | <i>Conventional<br/>Corporate Bond</i> | <i>Conventional<br/>Govt Bond</i> | <i>Islamic<br/>Corporate Bond</i> |
|-------------------------------------------------------------------------------|-------------------------------|--------------------------|--------------|----------------------------------------|-----------------------------------|-----------------------------------|
| <i>Panel A: Entire period, across all sample countries [annualized %]</i>     |                               |                          |              |                                        |                                   |                                   |
| Average                                                                       | 12.1342                       | 12.9918                  | 9.2140       | 8.4167                                 | 9.0156                            | 1.5338                            |
| Stdev                                                                         | 17.1069                       | 17.9080                  | 6.5127       | 24.7629                                | 7.1382                            | 4.5502                            |
| <i>Panel B: Entire period, world [annualized %]</i>                           |                               |                          |              |                                        |                                   |                                   |
| Average                                                                       | -3.3761                       | -2.9935                  |              | 0.5774                                 |                                   | 1.4411                            |
| Stdev                                                                         | 11.7910                       | 11.5512                  |              | 1.3582                                 |                                   | 6.7296                            |
| <i>Panel C: Non-crisis period, across all sample countries [annualized %]</i> |                               |                          |              |                                        |                                   |                                   |
| Average                                                                       | 14.1082                       | 17.6442                  | 11.4855      | 5.3423                                 | 13.0267                           | 2.8312                            |
| Stdev                                                                         | 12.8417                       | 13.6438                  | 4.4682       | 3.8954                                 | 4.9912                            | 1.9456                            |
| <i>Panel D: Non-crisis period, world [annualized %]</i>                       |                               |                          |              |                                        |                                   |                                   |
| Average                                                                       | -5.6104                       | -0.4777                  |              | 1.0714                                 |                                   | 0.8586                            |
| Stdev                                                                         | 6.8954                        | 7.0075                   |              | 1.3343                                 |                                   | 0.8261                            |
| <i>Panel E: Crisis period, across all sample countries [annualized %]</i>     |                               |                          |              |                                        |                                   |                                   |
| Average                                                                       | 11.4329                       | 11.3392                  | 8.4086       | 9.5018                                 | 7.5889                            | 1.0721                            |
| Stdev                                                                         | 18.3855                       | 19.1954                  | 7.0971       | 28.7103                                | 7.7581                            | 5.1701                            |
| <i>Panel F: Crisis period, world [annualized %]</i>                           |                               |                          |              |                                        |                                   |                                   |
| Average                                                                       | -3.3761                       | -2.9935                  |              | 0.5774                                 |                                   | 1.4411                            |
| Stdev                                                                         | 11.7910                       | 11.5512                  |              | 1.3582                                 |                                   | 6.7296                            |



**Table 8: Differences between Volatility Linkages that Involve Two Conventional Assets, and Volatility Linkages that Involve at Least One Islamic Asset for *GARCH*-Implied Volatilities**

This table reports in Panel A the results of testing for zero mean using five difference series of daily volatility linkages and all the difference series combined. The difference series of volatility linkages are obtained by subtracting the correlation that involves at least one Islamic (IS) asset from the corresponding correlation between conventional (C) assets. Assets include stock, government (Govt) bond, corporate (Corp) bond and money instruments. The correlations are estimated using Pearson correlation across *GARCH*-implied volatilities. We choose a *GARCH*(1,1) model for our return series based on the Akaike and the Schwarz Bayesian information criteria, and by examining the model residuals. In Panel B, the sample is split into Islamic and non-Islamic countries. Estimates that are significant at the 1%, 5% and 10% levels are denoted by superscripts a, b and c, respectively. Next to the t-statistics, these superscripts denote one-tailed tests, and next to the p-values, they refer to two-tailed tests.

*Panel A: Correlation of GARCH-implied Volatilities*

| <i>Differences</i> | <i>C Stock and C Govt Bond - IS Stock and C Govt Bond</i> | <i>C Stock and C Corp Bond - IS Stock and C Corp Bond</i> | <i>C Stock and C Money - IS Stock and C Money</i> | <i>C Stock and C Corp Bond - C Stock and IS Corp Bond</i> | <i>C Stock and C Corp Bond - IS Stock and IS Corp Bond</i> | <i>Total Differences</i> |
|--------------------|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|--------------------------|
| Observations       | 37                                                        | 27                                                        | 24                                                | 4                                                         | 4                                                          | 96                       |
| Mean               | 0.0288                                                    | 0.0231                                                    | 0.0203                                            | 0.1853                                                    | 0.2151                                                     | 0.0448                   |
| T-Statistic        | 1.5853                                                    | 2.3954 <sup>b</sup>                                       | 1.8607 <sup>c</sup>                               | 1.5319                                                    | 1.9043                                                     | 3.7133 <sup>a</sup>      |
| P-value            | 0.1216                                                    | 0.0241 <sup>b</sup>                                       | 0.0756 <sup>c</sup>                               | 0.2231                                                    | 0.1530                                                     | 0.0003 <sup>a</sup>      |

*Panel B: Differences by Country Type*

| <i>Differences</i> | <i>C Stock &amp; C Govt Bond - IS Stock &amp; C Govt Bond</i> |             | <i>C Stock &amp; C Corp Bond - IS Stock &amp; C Corp Bond</i> |             | <i>C Stock &amp; C Money - IS Stock &amp; C Money</i> |             | <i>Total Differences</i> |                     |
|--------------------|---------------------------------------------------------------|-------------|---------------------------------------------------------------|-------------|-------------------------------------------------------|-------------|--------------------------|---------------------|
| Sample             | Islamic                                                       | Non-Islamic | Islamic                                                       | Non-Islamic | Islamic                                               | Non-Islamic | Islamic                  | Non-Islamic         |
| Observations       | 3                                                             | 34          | 9                                                             | 18          | 4                                                     | 20          | 16                       | 72                  |
| Mean               | 0.0610                                                        | 0.0260      | 0.0399                                                        | 0.0147      | 0.0398                                                | 0.0164      | 0.0438                   | 0.0205              |
| T-Statistic        | 3.0017 <sup>b</sup>                                           | 1.3204      | 2.8973 <sup>a</sup>                                           | 1.1764      | 3.1839 <sup>a</sup>                                   | 1.2831      | 4.9134 <sup>a</sup>      | 1.9830 <sup>c</sup> |
| P-value            | 0.0954 <sup>b</sup>                                           | 0.1958      | 0.0200 <sup>b</sup>                                           | 0.2556      | 0.0499 <sup>a</sup>                                   | 0.2149      | 0.0002 <sup>a</sup>      | 0.0512 <sup>c</sup> |

*Test for equal means*

|             |        |        |        |        |
|-------------|--------|--------|--------|--------|
| Difference  | 0.0350 | 0.0252 | 0.0234 | 0.0234 |
| T-Statistic | 1.2386 | 1.3537 | 1.3123 | 1.7103 |
| P-value     | 0.2548 | 0.1908 | 0.2169 | 0.3002 |

## Appendix A

**Table A.1: Summary of the Data Availability**

The four categories below specify whether the countries used in this study are Islamic, and they list the specific indices that are available for these countries.

| <i>Categories</i>                                                                 |                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Islamic countries with Islamic stock and bond indices                          | Malaysia, United Arab Emirates (UAE) and Gulf Cooperation Council (GCC)                                                                                                                                                                                                                                                        |
| 2. Islamic countries with Islamic stock indices, but without Islamic bond indices | Indonesia, Pakistan, Qatar, Egypt, Morocco, Turkey                                                                                                                                                                                                                                                                             |
| 3. Other non-Islamic countries with Islamic stock indices                         | US, UK, Canada, Argentina, Australia, Austria, Belgium, Brazil, Chile, China, Colombia, Czech Republic, Denmark, Finland, France, Germany, Greece, Hungary, India, Ireland, Italy, Japan, Korea, Mexico, Netherlands, Norway, Peru, Philippines, Poland, Russia, Singapore, South Africa, Spain, Sweden, Switzerland, Thailand |
| 4. World index with Islamic stock and bond index                                  | World                                                                                                                                                                                                                                                                                                                          |

**Table A.2: List of Data Sources for the Stock Indices**

This table contains a complete list of the names, sources and time periods for the stock indices for all 46 countries in the study. Panel A lists all the stock indices used in the main analysis, and Panel B reports the stock indices used in the additional robustness tests.

| <i>Country</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <i>Index Name</i>                                                                                                                                                 | <i>Source and Period</i>                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <i>Panel A: Stock indices used for the main analysis</i>                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                   |                                                                           |
| Argentina, Australia, Austria, Belgium, Brazil, Canada, Chile, China, Colombia, Czech Republic, Denmark, Egypt, Finland, France, Gulf Cooperation Council (GCC), Germany, Greece, Hungary, India, Indonesia, Ireland, Italy, Japan, Malaysia, Mexico, Morocco, Netherlands, New Zealand, Norway, Pakistan, Peru, Philippines, Poland, Qatar, Russia, Singapore, South Africa, South Korea, Spain, Sweden, Switzerland, Thailand, Turkey, United Arab Emirates (UAE), UK, US, World | MSCI Standard Index and MSCI Standard Islamic Index                                                                                                               | 31 May 2007–8 June 2010, MSCI Barra [www.mscibarra.com/products/indices]  |
| <i>Panel B: Alternative stock indices used for robustness tests</i>                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                   |                                                                           |
| Canada, UK, US                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Dow Jones Islamic Index                                                                                                                                           | 30 December 1999–8 June 2010, Datastream                                  |
| Malaysia                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | FTSE Bursa Malaysia Hijrah Shariah Index                                                                                                                          | 31 May 2005–8 June 2010, Provided by FTSE Group Australia [info@ftse.com] |
| Australia, Canada, Egypt, Hungary, Indonesia, Malaysia, Pakistan, South Korea, UK, US                                                                                                                                                                                                                                                                                                                                                                                              | MSCI Large Cap Index, MSCI Large Cap Islamic Index, MSCI Mid Cap Index, MSCI Mid Cap Islamic Index, MSCI Small Cap Index, MSCI SMID Cap Index, MSCI IMI Cap Index | 31 May 2007–8 June 2010, MSCI Barra [www.mscibarra.com/products/indices]  |

**Table A.3: List of Data Sources for the Bond and Money Market Indices**

This is a complete list of the names, sources and time periods for the bond and money market indices for all 46 countries in the study. Panel A lists all the bond and money market indices used in the main analysis, and Panel B reports the bond and money market indices used in the additional robustness tests.

| Country                                                                                                                                                                                          | Index Name                                                                                                                  | Source and Period                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <i>Panel A: Bond and money market indices used for the main analysis</i>                                                                                                                         |                                                                                                                             |                                                                                                                    |
| Argentina, Brazil, Chile, China, Colombia, Czech Republic, Egypt, Hungary, India, Indonesia, Malaysia, Mexico, Peru, Philippines, Russia, Singapore, South Africa, South Korea, Thailand, Turkey | JPM ELMI+                                                                                                                   | 31 May 2007–8 June 2010, Datastream                                                                                |
| Argentina, Brazil, Colombia, Indonesia, Mexico, Peru, Philippines, Russia, South Africa, Turkey                                                                                                  | JPM EMBI+                                                                                                                   | 31 May 2007–8 June 2010, Datastream                                                                                |
| Chile, China, Egypt, Hungary, Pakistan                                                                                                                                                           | JPM EMBI Global                                                                                                             | 31 May 2007–8 June 2010, Datastream                                                                                |
| Argentina                                                                                                                                                                                        | JPM GBI-EM Broad Divers                                                                                                     | 31 May 2007–8 June 2010, Datastream                                                                                |
| Brazil, Malaysia                                                                                                                                                                                 | JPM GBI-EM Divers                                                                                                           | 31 May 2007–8 June 2010, Datastream                                                                                |
| China, India, Indonesia, Thailand                                                                                                                                                                | JPM GBI-EM Broad                                                                                                            | 31 May 2007–8 June 2010, Datastream                                                                                |
| Chile, Colombia, Peru, Russia, Turkey                                                                                                                                                            | JPM GBI-EM                                                                                                                  | 31 May 2007–8 June 2010, Datastream                                                                                |
| Czech Republic, Morocco, South Korea                                                                                                                                                             | JPM EURO EMBI Global                                                                                                        | 31 May 2007–8 June 2010, Datastream                                                                                |
| Austria, Belgium, Canada, Denmark, Finland, France, Greece, Ireland, Italy, Japan, Netherlands, New Zealand, Poland, Spain, Sweden, Switzerland, UK, US                                          | FTSE Global Government All Maturities                                                                                       | 31 May 2007–8 June 2010, Datastream                                                                                |
| Australia, New Zealand                                                                                                                                                                           | UBS Bank Bill All Maturities                                                                                                | 31 May 2007–8 June 2010, Datastream                                                                                |
| Australia                                                                                                                                                                                        | AFMA Australian Corporate Index, AFMA Australian Government Index                                                           | 31 May 2007–8 June 2010, Datastream                                                                                |
| Czech Republic, Germany                                                                                                                                                                          | Total All Lives DS Government Index                                                                                         | 31 May 2007–8 June 2010, Datastream                                                                                |
| Italy                                                                                                                                                                                            | Banca Fideuram Euro INV GDE Corporate Index                                                                                 | 31 May 2007–8 June 2010, Datastream                                                                                |
| Norway                                                                                                                                                                                           | Oslo Bors Government Fixed DM 0.25Y                                                                                         | 31 May 2007–8 June 2010, Datastream                                                                                |
| Singapore                                                                                                                                                                                        | Singapore Government Bond Overall                                                                                           | 31 May 2007–8 June 2010, Datastream                                                                                |
| South Africa                                                                                                                                                                                     | South African Government (GOVI)                                                                                             | 31 May 2007–8 June 2010, Datastream                                                                                |
| Qatar                                                                                                                                                                                            | Barclays Multiverse                                                                                                         | 31 May 2007–8 June 2010, Datastream                                                                                |
| World                                                                                                                                                                                            | Barclays Global Aggregate                                                                                                   | 31 May 2007–8 June 2010, Datastream                                                                                |
| UK                                                                                                                                                                                               | FTSE Sterling Corporate All Maturities                                                                                      | 31 May 2007–8 June 2010, Datastream                                                                                |
| US                                                                                                                                                                                               | Dow Jones Corporate Total Index                                                                                             | 31 May 2007–8 June 2010, Datastream                                                                                |
| World                                                                                                                                                                                            | Dow Jones Citigroup Sukuk Index                                                                                             | 31 May 2007–8 June 2010, Dow Jones Citigroup<br>[www.djindexes.com/fixedincome]                                    |
| Sweden                                                                                                                                                                                           | OM Benchmark Treasury Bills                                                                                                 | 31 May 2007–8 June 2010, Datastream                                                                                |
| Malaysia                                                                                                                                                                                         | Malaysian Corporate Bond Index, Malaysian Corporate Islamic Bond Index                                                      | 31 May 2007–8 June 2010, Quant Shop<br>[www.quantshop.com/malaysian%20bond%20v1.htm]                               |
| Gulf Cooperation Council (GCC) and United Arab Emirates (UAE)                                                                                                                                    | GCC Middle Eastern Conventional Bond Index, UAE Middle Eastern Conventional Bond Index, GCC Sukuk Index and UAE Sukuk Index | 31 May 2007–8 June 2010, Dubai International Financial Exchange,<br>[http://www.hsbc-nasdaqdubai.com/Default.aspx] |

**Table A.3 Continued***Panel B: Alternative bond and money market indices used for robustness tests*

|                                                               |                                                                                                 |                                                                                                                                                                              |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gulf Cooperation Council (GCC) and United Arab Emirates (UAE) | GCC Sukuk Index and UAE Sukuk Index                                                             | 31 May 2005–8 June 2010, Dubai International Financial Exchange, [ <a href="http://www.hsbc-nasdaqdubai.com/Default.aspx">http://www.hsbc-nasdaqdubai.com/Default.aspx</a> ] |
| Canada, UK, US                                                | FTSE Global Government All Maturities                                                           | 31 May 2005–8 June 2010, Datastream                                                                                                                                          |
| Malaysia                                                      | JPM GBI-EM Divers, JPM EMBI Global                                                              | 31 May 2005–8 June 2010, Datastream                                                                                                                                          |
| Australia                                                     | UBS Treasury All Maturities, Macquarie BK Corporate Bond, FTSE Global Government All Maturities | 31 May 2007–8 June 2010, Datastream                                                                                                                                          |
| Hungary                                                       | Hungary Government DMA MAX Bill Index                                                           | 31 May 2007–8 June 2010, Datastream                                                                                                                                          |
| Indonesia, Pakistan                                           | Barclays EM Asia                                                                                | 31 May 2007–8 June 2010, Datastream                                                                                                                                          |
| South Korea                                                   | HSBC ADBI Korea                                                                                 | 31 May 2007–8 June 2010, Datastream                                                                                                                                          |
| Malaysia                                                      | JPM EMBI Global                                                                                 | 31 May 2007–8 June 2010, Datastream                                                                                                                                          |
| Malaysia                                                      | Malaysian Government Bond Index                                                                 | 31 May 2007–8 June 2010, Quant Shop [ <a href="http://www.quantshop.com/malaysian%20bond%20v1.htm">www.quantshop.com/malaysian%20bond%20v1.htm</a> ]                         |

**Table A.4: List of Control Variables and their Sources**

This table presents a complete list of the control variables and their specifications including time and frequency, source, country or asset level and further calculations.

| <i>Control variable</i>                                                                                                                                                                                                                                                                                                                                                     | <i>Frequency</i>    | <i>Source</i>                                       | <i>Country or asset level</i> | <i>Further calculations</i>                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sovereign credit ratings (long term and short term)                                                                                                                                                                                                                                                                                                                         | annual frequency    | Fitch credit ratings                                | Country                       |                                                                                                                                                                                                                                                                                                    |
| Country risk classification                                                                                                                                                                                                                                                                                                                                                 | quarterly frequency | OECD                                                | Country                       |                                                                                                                                                                                                                                                                                                    |
| Periods of high (low) volatility                                                                                                                                                                                                                                                                                                                                            | monthly frequency   | Estimated from the original spot and futures series | Asset                         | Two indicator variables are calculated that take the value of 1 if the volatility of at least one of the two assets for which the volatility linkage will be estimated falls within the upper (lower) 25% of the volatility distribution where monthly squared returns is the volatility proxy     |
| GDP per capita: seasonally adjusted nominal GDP and population                                                                                                                                                                                                                                                                                                              | quarterly frequency | Datastream                                          | Country                       | Seasonally adjusted nominal GDP divided by total population                                                                                                                                                                                                                                        |
| 'Ease of doing business' index: the enforcement of contracts (measured either as procedures required, time or cost), the total tax rate, the strength of legal rights index, <sup>26</sup> the extent of disclosure index, the strength of investor protection index, <sup>27</sup> private credit bureau coverage, <sup>28</sup> and the depth of credit information index | annual frequency    | World Bank                                          | Country                       |                                                                                                                                                                                                                                                                                                    |
| Corruption index                                                                                                                                                                                                                                                                                                                                                            | annual frequency    | Transparency International                          | Country                       |                                                                                                                                                                                                                                                                                                    |
| Foreign direct investment                                                                                                                                                                                                                                                                                                                                                   | annual frequency    | World Bank                                          | Country                       |                                                                                                                                                                                                                                                                                                    |
| Average effective bid-ask spread                                                                                                                                                                                                                                                                                                                                            | monthly frequency   | Estimated from the original spot and futures series | Asset                         | The effective bid-ask spread is measured by $spread = 2 \times \sqrt{ Cov(r_t, r_{t-1}) }$ where the covariance refers to the first-order serial covariance of price returns. The spread is calculated in monthly intervals and then the average is computed for the spread of each asset pairing. |
| Corporate bond dummy and corporate money market dummy                                                                                                                                                                                                                                                                                                                       | monthly frequency   |                                                     | Asset                         | For bond indices, the corporate dummy variable equals 1 with a corporate bond index and 0 with a government bond index. For money market indices, the variable equals 1 when it refers to Emerging Local Markets Index (ELMI) <sup>29</sup> and 0 when it includes pure Treasury or bank bills.    |

<sup>26</sup> The strength of legal rights index measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders and thus facilitate lending.

<sup>27</sup> The strength of investor protection index is the average of the extent of disclosure index, the extent of director liability index and the ease of shareholder suits index.

<sup>28</sup> Private credit bureau coverage reports the number of individuals or firms listed by a private credit bureau with current information on repayment history, unpaid debts or credit outstanding.

<sup>29</sup> The ELMI indices include a mixture of local-currency-denominated money market instruments, such as treasury bills, FX forwards and deposits with maturities ranging from 1 to 13 weeks.

## Appendix B

**Table B.1: Descriptive Statistics for a Sub-Sample of Countries**

This table provides the annualized returns and standard deviation (in US dollars) for 13 countries. The annualized values are obtained by multiplying the daily returns and standard deviation by the average trading days per year.

| <i>Country</i> | <i>Observations</i> | <i>Average trading days per year</i> | <i>Contract</i>              | <i>Annualized Return</i> | <i>Annualized Standard Deviation</i> |
|----------------|---------------------|--------------------------------------|------------------------------|--------------------------|--------------------------------------|
| Australia      | 765                 | 254                                  | Islamic Stock                | -0.014                   | 3.092                                |
|                |                     |                                      | Conventional Stock           | -0.030                   | 2.804                                |
|                |                     |                                      | Conventional Corporate Bond  | 0.027                    | 1.293                                |
|                |                     |                                      | Conventional Government Bond | 0.030                    | 1.252                                |
|                |                     |                                      | Conventional Money           | 0.023                    | 1.357                                |
| GCC            | 741                 | 243.5                                | Islamic Stock                | -0.061                   | 1.852                                |
|                |                     |                                      | Conventional Stock           | -0.045                   | 1.734                                |
|                |                     |                                      | Islamic Corporate Bond       | 0.003                    | 0.651                                |
|                |                     |                                      | Conventional Corporate Bond  | 0.013                    | 0.415                                |
| Hungary        | 764                 | 254                                  | Islamic Stock                | -0.059                   | 3.501                                |
|                |                     |                                      | Conventional Stock           | -0.087                   | 3.774                                |
|                |                     |                                      | Conventional Corporate Bond  | 0.015                    | 0.953                                |
|                |                     |                                      | Conventional Government Bond | -0.004                   | 1.761                                |
|                |                     |                                      | Conventional Money           | -0.001                   | 1.426                                |
| Indonesia      | 738                 | 244.5                                | Islamic Stock                | 0.029                    | 3.019                                |
|                |                     |                                      | Conventional Stock           | 0.056                    | 2.802                                |
|                |                     |                                      | Conventional Corporate Bond  | 0.033                    | 2.031                                |
|                |                     |                                      | Conventional Government Bond | 0.039                    | 1.719                                |
|                |                     |                                      | Conventional Money           | 0.047                    | 1.040                                |
| Japan          | 744                 | 246.5                                | Islamic Stock                | -0.044                   | 2.047                                |
|                |                     |                                      | Conventional Stock           | -0.056                   | 1.997                                |
|                |                     |                                      | Conventional Government Bond | 0.056                    | 0.976                                |
| Malaysia       | 748                 | 247                                  | Islamic Stock                | 0.005                    | 1.546                                |
|                |                     |                                      | Conventional Stock           | 0.005                    | 1.477                                |
|                |                     |                                      | Conventional Corporate Bond  | 0.021                    | 0.536                                |
|                |                     |                                      | Islamic Corporate Bond       | 0.021                    | 0.532                                |
|                |                     |                                      | Conventional Government Bond | 0.017                    | 0.558                                |
|                |                     |                                      | Conventional Money           | 0.015                    | 0.532                                |
| Morocco        | 744                 | 247                                  | Islamic Stock                | 0.003                    | 1.480                                |
|                |                     |                                      | Conventional Stock           | 0.000                    | 1.437                                |
|                |                     |                                      | Conventional Corporate Bond  | 0.012                    | 1.163                                |
| Pakistan       | 752                 | 249                                  | Islamic Stock                | -0.059                   | 2.301                                |
|                |                     |                                      | Conventional Stock           | -0.092                   | 2.243                                |
|                |                     |                                      | Conventional Corporate Bond  | 0.036                    | 1.969                                |
| Qatar          | 766                 | 253                                  | Islamic Stock                | 0.011                    | 2.470                                |
|                |                     |                                      | Conventional Stock           | 0.015                    | 2.311                                |
|                |                     |                                      | Conventional Corporate Bond  | 104.235                  | 416.903                              |
| Turkey         | 772                 | 256                                  | Islamic Stock                | -0.037                   | 3.219                                |
|                |                     |                                      | Conventional Stock           | -0.007                   | 3.385                                |

|     |     |       |                              |        |       |
|-----|-----|-------|------------------------------|--------|-------|
| UAE | 738 | 242   | Conventional Corporate Bond  | 0.038  | 0.939 |
|     |     |       | Conventional Government Bond | 0.054  | 1.492 |
|     |     |       | Conventional Money           | 0.031  | 1.238 |
|     |     |       | Islamic Stock                | -0.172 | 3.180 |
|     |     |       | Conventional Stock           | -0.109 | 2.534 |
|     |     |       | Islamic Corporate Bond       | 0.009  | 0.833 |
| UK  | 770 | 255.5 | Conventional Corporate Bond  | 0.008  | 0.494 |
|     |     |       | Islamic Stock                | -0.042 | 2.461 |
|     |     |       | Conventional Stock           | -0.069 | 2.433 |
|     |     |       | Conventional Corporate Bond  | -0.011 | 0.932 |
| US  | 758 | 251.5 | Conventional Government Bond | -0.012 | 0.955 |
|     |     |       | Islamic Stock                | -0.023 | 1.904 |
|     |     |       | Conventional Stock           | -0.042 | 2.100 |
|     |     |       | Conventional Corporate Bond  | 0.036  | 0.535 |
|     |     |       | Conventional Government Bond | 0.031  | 0.374 |



**Table B.2: Estimated Volatility Linkages for a Sub-Sample of Countries**

This table summarizes the volatility correlations across the whole sample period from 31 May 2007 to 8 June 2010 for 13 countries. In Panel A the volatility correlations are computed as the Pearson correlation between the daily absolute return series of two assets. In Panel B the volatility correlations are computed as the Pearson correlation between the daily squared return series of two assets. Panel C provides the *GMM* estimates for the volatility correlations that are computed using Fleming, Kirby, and Ostdiek's (1998) stochastic volatility model and the arising moment conditions in Eq. (20). In Panel C the volatility correlations are only reported when the overall *GMM* model cannot be rejected at the 10% level of significance. The lag length in all models is  $L = 40$ . Furthermore, in all panels the volatility correlations are computed between 11 combinations of Islamic (IS) and conventional (C) stock (Stock), government bond (Govt Bond), corporate bond (Corp Bond) and money market (Money) indices.

| Asset Pairing                             | Stock and Bond           |                          |                           |                         |                           |                            | Stock and Money      |                       |
|-------------------------------------------|--------------------------|--------------------------|---------------------------|-------------------------|---------------------------|----------------------------|----------------------|-----------------------|
| Panel A: Correlations of Absolute Returns |                          |                          |                           |                         |                           |                            |                      |                       |
| Assets                                    | C Stock –<br>C Govt Bond | C Stock –<br>C Corp Bond | IS Stock –<br>C Govt Bond | IS Stock-C<br>Corp Bond | C Stock –<br>IS Corp Bond | IS Stock - IS<br>Corp Bond | C Stock –<br>C Money | IS Stock –<br>C Money |
| Australia                                 | 0.5277                   | 0.5520                   | 0.5223                    | 0.5516                  |                           |                            | 0.5725               | 0.5727                |
| GCC                                       |                          | 0.1324                   |                           | 0.1287                  | 0.0978                    | 0.0898                     |                      |                       |
| Hungary                                   | 0.6843                   | 0.3339                   | 0.5764                    | 0.3087                  |                           |                            | 0.6370               | 0.5554                |
| Indonesia                                 | 0.6091                   | 0.3039                   | 0.5025                    | 0.2180                  |                           |                            | 0.5547               | 0.4555                |
| Japan                                     | 0.1579                   |                          | 0.1607                    |                         |                           |                            |                      |                       |
| Malaysia                                  | 0.4055                   | 0.3699                   | 0.3467                    | 0.3098                  | 0.3773                    | 0.3181                     | 0.4394               | 0.4098                |
| Morocco                                   |                          | 0.3767                   |                           | 0.2862                  |                           |                            |                      |                       |
| Pakistan                                  |                          | 0.0581                   |                           | 0.0442                  |                           |                            |                      |                       |
| Qatar                                     |                          | 0.1394                   |                           | 0.1348                  |                           |                            |                      |                       |
| Turkey                                    | 0.7480                   | 0.4060                   | 0.7013                    | 0.3948                  |                           |                            | 0.6692               | 0.6376                |
| UAE                                       |                          | 0.3305                   |                           | 0.2926                  | 0.2795                    | 0.2459                     |                      |                       |
| UK                                        | 0.3490                   | 0.4910                   | 0.3423                    | 0.4860                  |                           |                            |                      |                       |
| USA                                       | 0.2734                   | 0.3140                   | 0.2647                    | 0.3056                  |                           |                            |                      |                       |

| <b>Panel B: Correlations of Squared Returns</b> |                          |                          |                           |                           |                           |                            |                      |                       |
|-------------------------------------------------|--------------------------|--------------------------|---------------------------|---------------------------|---------------------------|----------------------------|----------------------|-----------------------|
| Assets                                          | C Stock – C<br>Govt Bond | C Stock - C<br>Corp Bond | IS Stock - C<br>Govt Bond | IS Stock - C<br>Corp Bond | C Stock - IS<br>Corp Bond | IS Stock - IS<br>Corp Bond | C Stock - C<br>Money | IS Stock - C<br>Money |
| Australia                                       | 0.4507                   | 0.4717                   | 0.4546                    | 0.4802                    |                           |                            | 0.3305               | 0.3180                |
| GCC                                             |                          | 0.0298                   |                           | 0.0264                    | 0.0053                    | 0.0023                     |                      |                       |
| Hungary                                         | 0.7544                   | 0.4225                   | 0.6941                    | 0.3980                    |                           |                            | 0.7186               | 0.6492                |
| Indonesia                                       | 0.6884                   | 0.1945                   | 0.6164                    | 0.1260                    |                           |                            | 0.5892               | 0.5353                |
| Japan                                           | 0.1764                   |                          | 0.1925                    |                           |                           |                            |                      |                       |
| Malaysia                                        | 0.2843                   | 0.2237                   | 0.2052                    | 0.2586                    | 0.2238                    | 0.2062                     | 0.2974               | 0.3233                |
| Morocco                                         |                          | 0.3087                   |                           | 0.2303                    |                           |                            |                      |                       |
| Pakistan                                        |                          | 0.7415                   |                           | 0.7382                    |                           |                            |                      |                       |
| Qatar                                           |                          | 0.0490                   |                           | 0.0457                    |                           |                            |                      |                       |
| Turkey                                          | 0.7748                   | 0.2623                   | 0.7439                    | 0.2235                    |                           |                            | 0.6637               | 0.6543                |
| UAE                                             |                          | 0.3131                   |                           | 0.2412                    | 0.2525                    | 0.1756                     |                      |                       |
| UK                                              | 0.3439                   | 0.2181                   | 0.3392                    | 0.2326                    |                           |                            |                      |                       |
| USA                                             | 0.2641                   | 0.9815                   | 0.2494                    | 0.9958                    |                           |                            |                      |                       |

**Panel C: GMM Estimates**

| Assets    | C Stock –<br>C Govt Bond | C Stock –<br>C Corp Bond | IS Stock –<br>C Govt Bond | IS Stock –<br>C Corp Bond | C Stock - IS<br>Corp Bond | IS Stock - IS<br>Corp Bond | C Stock - C<br>Money | IS Stock - C<br>Money |
|-----------|--------------------------|--------------------------|---------------------------|---------------------------|---------------------------|----------------------------|----------------------|-----------------------|
| Australia |                          | 0.4126                   |                           |                           |                           |                            |                      |                       |
| GCC       |                          | 0.2935                   |                           | 0.2652                    | 0.2476                    | 0.2331                     |                      |                       |
| Hungary   |                          |                          |                           |                           |                           |                            | 0.5315               |                       |
| Indonesia | 0.6901                   | 0.5365                   | 0.6241                    | 0.4059                    |                           |                            |                      | 0.4486                |
| Japan     | -0.2902                  |                          |                           |                           |                           |                            |                      |                       |
| Malaysia  | 0.1538                   | 0.0000                   |                           | -0.2397                   | 0.0000                    | -0.1999                    | 0.1548               | 0.0000                |
| Morocco   |                          | 0.5843                   |                           | 0.4580                    |                           |                            |                      |                       |
| Pakistan  |                          | 0.0000                   |                           | -0.1669                   |                           |                            |                      |                       |
| Qatar     |                          | 0.1480                   |                           |                           |                           |                            |                      |                       |
| Turkey    |                          |                          |                           | 0.4795                    |                           |                            |                      | 0.3199                |
| UAE       |                          | 0.2534                   |                           | 0.1645                    | 0.3292                    |                            |                      |                       |
| UK        | 0.1170                   | 0.1858                   |                           |                           |                           |                            |                      |                       |
| USA       | 0.6070                   | 0.8092                   |                           | 0.5103                    |                           |                            |                      |                       |

**Table B.3: Sensitivity of Volatility Linkage Estimates to Alternative Sample Periods using Pearson Correlation and the Absolute Return Volatility Proxy for a Sub-Sample of Countries**

This table presents additional robustness results (with volatility linkages computed using Pearson correlation across absolute returns) for extended periods of time. The volatility linkages for the six countries below involve Islamic (IS) and conventional stock, corporate bond (Corp Bond) and government bond (Govt Bond) markets. The sample periods considered are: 31 December 1999 to 8 June 2010 (Panel A), 31 May 2005 to 8 June 2010 (Panel B), and 31 May 2007 to 8 June 2010 (Panel C).

| Country                                                 | Canada | GCC    | Malaysia | UAE    | UK     | USA    |
|---------------------------------------------------------|--------|--------|----------|--------|--------|--------|
| <i>Panel A: Years 2000 – 2010</i>                       |        |        |          |        |        |        |
| C Stock & C Govt Bond                                   | 0.3831 |        |          |        | 0.2074 | 0.2258 |
| IS Stock & C Govt Bond                                  | 0.3062 |        |          |        | 0.1940 | 0.1901 |
| <i>Panel B: Years 2005 – 2010</i>                       |        |        |          |        |        |        |
| C Stock & C Govt Bond                                   | 0.4979 |        | 0.4174   |        | 0.2738 | 0.3172 |
| IS Stock & C Govt Bond                                  | 0.4689 |        | 0.3609   |        | 0.2577 | 0.3039 |
| C Stock & C Corp Bond                                   |        |        |          |        |        |        |
| IS Stock & C Corp Bond                                  |        |        |          |        |        |        |
| C Stock & C Corp Bond                                   |        | 0.0853 |          | 0.2874 |        |        |
| C Stock & IS Corp Bond                                  |        | 0.0698 |          | 0.2524 |        |        |
| <i>Panel C: Main Analysis Sample: Years 2007 - 2010</i> |        |        |          |        |        |        |
| C Stock & C Govt Bond                                   | 0.4806 |        | 0.4020   |        | 0.2378 | 0.2734 |
| IS Stock & C Govt Bond                                  | 0.4642 |        | 0.3446   |        | 0.2236 | 0.2666 |
| C Stock & C Corp Bond                                   |        |        |          |        |        |        |
| IS Stock & C Corp Bond                                  |        |        |          |        |        |        |
| C Stock & C Corp Bond                                   |        | 0.1324 |          | 0.3305 |        |        |
| C Stock & IS Corp Bond                                  |        | 0.0978 |          | 0.2795 |        |        |

**Table B.4: Sensitivity of Volatility Linkage Estimates to Alternative Bond and Money Market Indices using Pearson Correlation and the Absolute Return Volatility Proxy for a Sub-Sample of Countries**

This table reports the results for the second additional robustness test for which volatility linkages are computed using Pearson correlation across absolute returns. The robustness test involves estimating the volatility linkages using alternative asset market proxies (Panel A) for five countries. For comparison purposes, the previous estimates using the original asset market proxies are presented in Panel B. The volatility linkages involve Islamic (IS) and conventional (C) stock, corporate bond (Corp Bond), government bond (Govt Bond) and money markets, and the sample period is from 31 May 2007 to 8 June 2010.

| <i>Country</i>                                      | <i>Australia</i> | <i>Hungary</i> | <i>Indonesia</i> | <i>Malaysia</i> | <i>Pakistan</i> | <i>South Korea</i> |
|-----------------------------------------------------|------------------|----------------|------------------|-----------------|-----------------|--------------------|
| <i>Panel A: New Asset Market Proxies</i>            |                  |                |                  |                 |                 |                    |
| C Stock & C Govt Bond                               | 0.5041           |                |                  | 0.4055          |                 |                    |
| IS Stock & C Govt Bond                              | 0.5054           |                |                  | 0.3467          |                 |                    |
| C Stock & Money                                     | 0.5168           | 0.6741         |                  |                 |                 |                    |
| IS Stock & Money                                    | 0.5124           | 0.5907         |                  |                 |                 |                    |
| C Stock & C Corp Bond                               | 0.5727           |                | 0.1828           | 0.1693          | 0.0750          | 0.2491             |
| IS Stock & C Corp Bond                              | 0.5719           |                | 0.0860           | 0.1907          | 0.0774          | 0.2511             |
| <i>Panel B: Main Analysis Sample Market Proxies</i> |                  |                |                  |                 |                 |                    |
| C Stock & C Govt Bond                               | 0.5277           |                |                  | 0.4055          |                 |                    |
| IS Stock & C Govt Bond                              | 0.5223           |                |                  | 0.3467          |                 |                    |
| C Stock & Money                                     | 0.5725           | 0.6370         |                  |                 |                 |                    |
| IS Stock & Money                                    | 0.5727           | 0.5554         |                  |                 |                 |                    |
| C Stock & C Corp Bond                               | 0.5520           |                | 0.3039           | 0.3699          | 0.0581          | 0.1658             |
| IS Stock & C Corp Bond                              | 0.5516           |                | 0.2180           | 0.3098          | 0.0442          | 0.1655             |

**Table B.5: Sensitivity of Volatility Linkage Estimates to Alternative Stock Indices using Pearson Correlation and the Absolute Return Volatility Proxy for a Sub-Sample of Countries**

This table summarises the results for the third additional robustness test, where volatility linkages are computed using Pearson correlation across absolute returns. The robustness test involves estimating the volatility linkages between alternative stock market indices and a money market index (Panel A), a conventional corporate bond market index (Panel B), a conventional government bond market index (Panel C) and an Islamic corporate bond market index (Panel D) for ten countries. The various stock market indices refer to conventional (C) and Islamic (IS) indices with various market capitalizations, including large, mid, small, SMID (small and mid) and IMI (small, mid and large). The sample period is from 31 May 2007 to 8 June 2010.

| Country                                                 | Australia | Canada | Egypt  | Hungary | Indonesia | Malaysia | Pakistan | South Korea | UK | USA    |
|---------------------------------------------------------|-----------|--------|--------|---------|-----------|----------|----------|-------------|----|--------|
| <i>Panel A: Stock and Money Market</i>                  |           |        |        |         |           |          |          |             |    |        |
| C Large Stock                                           | 0.5624    |        | 0.0406 | 0.6293  | 0.5518    | 0.3983   |          | 0.7311      |    |        |
| IS Large Stock                                          | 0.5527    |        | 0.0572 |         | 0.4374    | 0.3694   |          |             |    |        |
| C Mid Stock                                             | 0.5821    |        | 0.0418 | 0.5648  | 0.5059    | 0.3939   |          | 0.7168      |    |        |
| IS Mid Stock                                            | 0.5691    |        | 0.0548 |         | 0.4947    | 0.3130   |          |             |    |        |
| C Small Stock                                           | 0.6126    |        | 0.0438 | 0.6342  | 0.4713    | 0.4184   |          | 0.7331      |    |        |
| C SMID Stock                                            | 0.6025    |        | 0.0495 | 0.6008  | 0.5163    | 0.4287   |          | 0.7299      |    |        |
| C IMI Stock                                             |           |        | 0.0486 | 0.6405  | 0.5555    | 0.4247   |          | 0.7380      |    |        |
| <i>Panel B: Stock and Corporate Bond Market</i>         |           |        |        |         |           |          |          |             |    |        |
| C Large Stock                                           | 0.5416    |        | 0.1682 | 0.3501  | 0.3006    | 0.3377   | 0.0587   | 0.1624      |    | 0.3170 |
| IS Large Stock                                          | 0.5318    |        | 0.1377 |         | 0.2095    | 0.2670   | 0.0517   |             |    | 0.3046 |
| C Mid Stock                                             | 0.5640    |        | 0.1282 | 0.2615  | 0.2965    | 0.3215   | 0.0401   | 0.1681      |    | 0.2932 |
| IS Mid Stock                                            | 0.5487    |        | 0.0778 |         | 0.3119    | 0.2422   | 0.0503   |             |    | 0.2801 |
| C Small Stock                                           | 0.5932    |        | 0.2328 | 0.2735  | 0.2540    | 0.3308   |          | 0.1771      |    | 0.2841 |
| C SMID Stock                                            | 0.5836    |        | 0.2030 | 0.2674  | 0.2871    | 0.3428   |          | 0.1744      |    | 0.2823 |
| C IMI Stock                                             |           |        | 0.1903 | 0.3302  | 0.3070    | 0.3534   |          | 0.1630      |    |        |
| <i>Panel C: Stock and Government Bond Mark</i>          |           |        |        |         |           |          |          |             |    |        |
| C Large Stock                                           |           |        | 0.5179 | 0.4799  |           |          |          | 0.5926      |    | 0.3556 |
| IS Large Stock                                          |           |        | 0.5024 | 0.4517  |           |          |          | 0.4751      |    | 0.2853 |
| C Mid Stock                                             |           |        | 0.5375 | 0.4826  |           |          |          | 0.5628      |    | 0.3608 |
| IS Mid Stock                                            |           |        | 0.5228 | 0.4608  |           |          |          | 0.5288      |    | 0.2682 |
| C Small Stock                                           |           |        | 0.5678 | 0.5189  |           |          |          | 0.5265      |    | 0.3590 |
| C SMID Stock                                            |           |        | 0.5569 | 0.5027  |           |          |          | 0.5737      |    | 0.3800 |
| C IMI Stock                                             |           |        |        |         |           |          |          | 0.6016      |    | 0.3813 |
| <i>Panel D: Stock and Islamic Corporate Bond Market</i> |           |        |        |         |           |          |          |             |    |        |
| C Large Stock                                           |           |        |        |         | 0.3443    |          |          |             |    |        |
| IS Large Stock                                          |           |        |        |         | 0.2747    |          |          |             |    |        |
| C Mid Stock                                             |           |        |        |         | 0.3288    |          |          |             |    |        |
| IS Mid Stock                                            |           |        |        |         | 0.2485    |          |          |             |    |        |
| C Small Stock                                           |           |        |        |         | 0.3402    |          |          |             |    |        |
| C SMID Stock                                            |           |        |        |         | 0.3512    |          |          |             |    |        |
| C IMI Stock                                             |           |        |        |         | 0.3606    |          |          |             |    |        |